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# **Software Requirements Specification**

**for**

# **Cyber Sentinel AI System**

**Version 1.0 approved**

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## Revision History

| Name | Date | Reason For Changes | Version |
|------|------|--------------------|---------|
|      |      |                    |         |
|      |      |                    |         |

## 1. Introduction

### 1.1 Purpose

*This Software Requirements Specification (SRS) document describes the requirements for the Cyber Sentinel AI System Version 1.0. The purpose of this document is to define the functional and non-functional requirements of the system and provide a clear understanding of its features, behavior, and constraints. The Cyber Sentinel AI System is designed to detect, analyze, report, and manage cyber threats such as phishing attacks, malicious URLs, scam messages, suspicious emails, and fraudulent QR codes. This SRS serves as a reference for developers, testers, project managers, and stakeholders involved in the design, implementation, deployment, and maintenance of the system.*

### 1.2 Document Conventions

*This document follows the IEEE Software Requirements Specification (SRS) standard. Section headings and subheadings are numbered hierarchically for easy navigation. Functional requirements are described using clear and concise statements. Important terms and system components are written in bold when first introduced. Requirements are categorized into functional and non-functional requirements. All requirement statements are assigned individual priorities based on their importance and implementation needs. Diagrams, tables, and figures are referenced using appropriate numbering conventions throughout the document.*

### 1.3 Intended Audience and Reading Suggestions

*This document is intended for software developers, cybersecurity analysts, testers, project managers, system administrators, stakeholders, and end users. Developers can use this document to understand system functionalities and technical requirements. Testers can refer to the requirements for creating test cases and validating system behavior. Project managers can use the document to monitor project scope and progress. System administrators can understand deployment and maintenance requirements, while stakeholders can review the overall objectives and capabilities of the system. Readers are advised to begin with the Introduction and Overall Description sections before proceeding to the Functional Requirements, System Features, and Non-Functional Requirements sections relevant to their role.*

## 1.4 Product Scope

*The Cyber Sentinel AI System is an intelligent cybersecurity platform developed to help individuals, students, senior citizens, businesses, and organizations identify and respond to cyber threats effectively. The system utilizes artificial intelligence techniques to analyze URLs, QR codes, emails, and suspicious messages for detecting scams, phishing attacks, malware distribution attempts, and other malicious activities. Users can report incidents anonymously, upload evidence, track case status, and receive risk assessments with recommended actions. The platform also provides community scam alerts, real-time threat analytics, incident management, and investigation support. The primary objective of the system is to improve cyber awareness, enhance threat detection capabilities, and provide a centralized platform for cyber incident reporting and response management.*

## 1.5 References

*The following references were used during the preparation of this Software Requirements Specification:*

- 1. IEEE Recommended Practice for Software Requirements Specifications (IEEE Std 830-1998).*
- 2. IEEE Software and Systems Engineering Standards Collection.*
- 3. OWASP Web Security Testing Guide, Latest Version.*
- 4. National Institute of Standards and Technology (NIST) Cybersecurity Framework.*
- 5. Cybersecurity and Infrastructure Security Agency (CISA) Guidelines and Best Practices.*
- 6. Official Documentation of HTML5, CSS3, JavaScript, and Node.js.*
- 7. Database Documentation for MySQL/MongoDB.*
- 8. Artificial Intelligence and Machine Learning Documentation relevant to phishing and scam detection.*
- 9. Project Design Documents, Use Case Specifications, and System Architecture Documents developed for the Cyber Sentinel AI System.*

## 2. Overall Description

### 2.1 Product Perspective

*The Cyber Sentinel AI System is a standalone web-based cybersecurity platform developed to assist users in identifying, reporting, and managing cyber threats. The system is designed as a centralized solution that combines artificial intelligence, incident reporting, threat analysis, and cybersecurity awareness features. It serves as a proactive defense mechanism against phishing attacks, scam messages, malicious URLs, fraudulent QR codes, and other cyber risks. The platform integrates user reporting, evidence management, threat analytics, and administrative investigation modules into a single environment. It can also be extended in the future to integrate with external cybersecurity databases, threat intelligence feeds, and law enforcement reporting systems.*

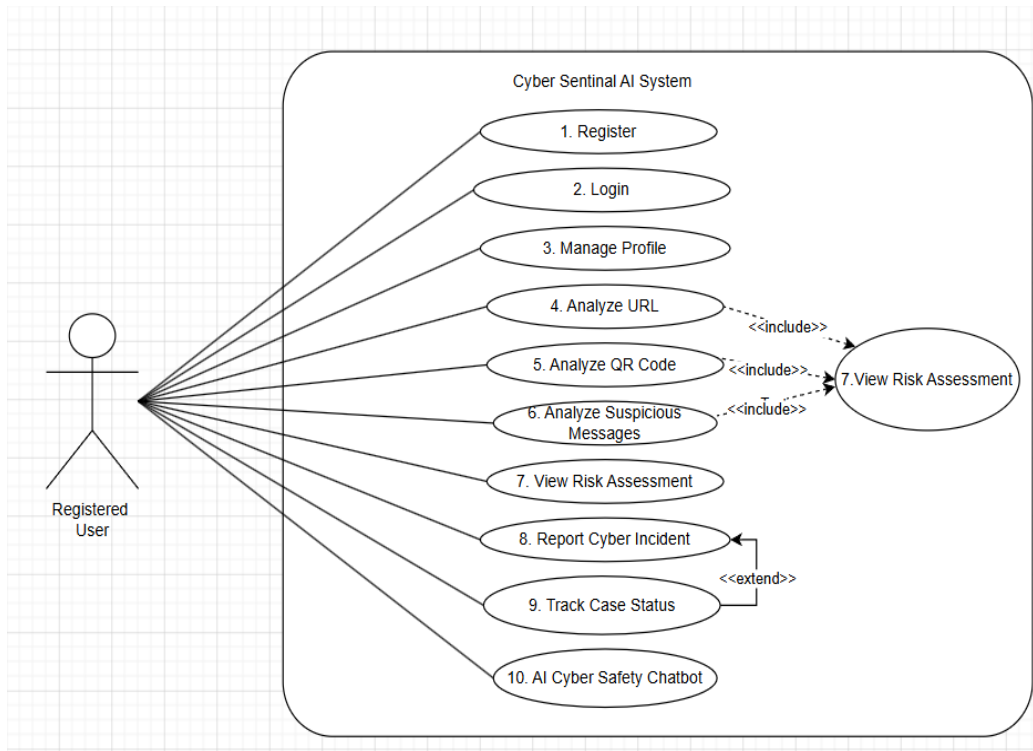
### 2.2 Product Functions

*The Cyber Sentinel AI System provides the following major functions:*

- *User registration and secure authentication.*
- *AI-powered analysis of URLs, QR codes, emails, and suspicious messages.*
- *Detection of phishing attempts, scams, and malicious content.*
- *Anonymous cyber incident reporting.*
- *Evidence upload and management.*
- *Incident status tracking and case management.*
- *Risk assessment and security recommendations.*
- *Community scam alerts and notifications.*
- *Threat analytics and visualization dashboard.*
- *Incident assignment and investigation management.*
- *Administrative monitoring and report generation.*
- *Cybersecurity awareness and educational resources.*
- *User feedback and support services.*

## 2.3 User Classes and Characteristics

### 1.Registered User:



#### Use Case Description

**Actor:** Registered User

**Use Case:** Analyze URL

**Primary Actor:** Registered User

**Secondary Actor:** AI Threat Detection System

**Precondition:**

- User must be logged into the system.

**Postcondition:**

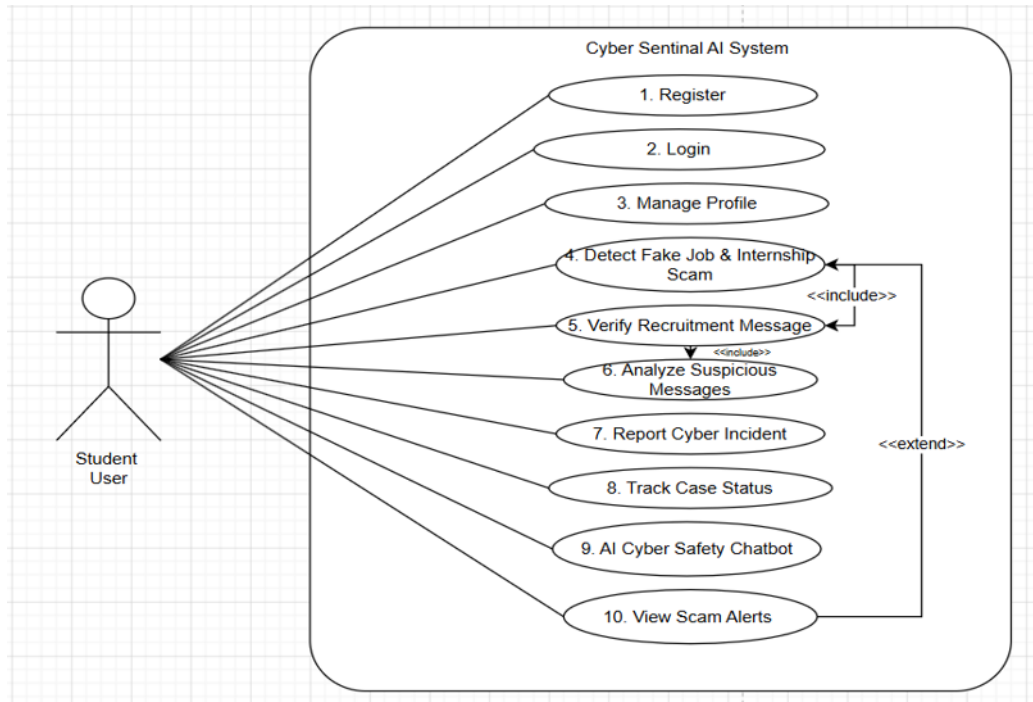
- URL is analyzed and the risk level is displayed.

#### Main Success Scenario

1. User enters a website URL.
2. System validates the URL.
3. AI analyzes the URL.
4. System identifies whether it is safe or malicious.
5. Risk assessment is generated.
6. Result is displayed to the user.

**Exception Scenario**

- If the URL is invalid, the system displays an error message.
- If the analysis service is unavailable, the user is asked to try again later.

**2.Student User:****Use Case Description****Actor:** Student User**Use Case:** Detect Fake Job & Internship Scam**Primary Actor:** Student User**Secondary Actor:** AI Scam Detection System**Precondition:**

- Student must be logged into the system.

**Postcondition:**

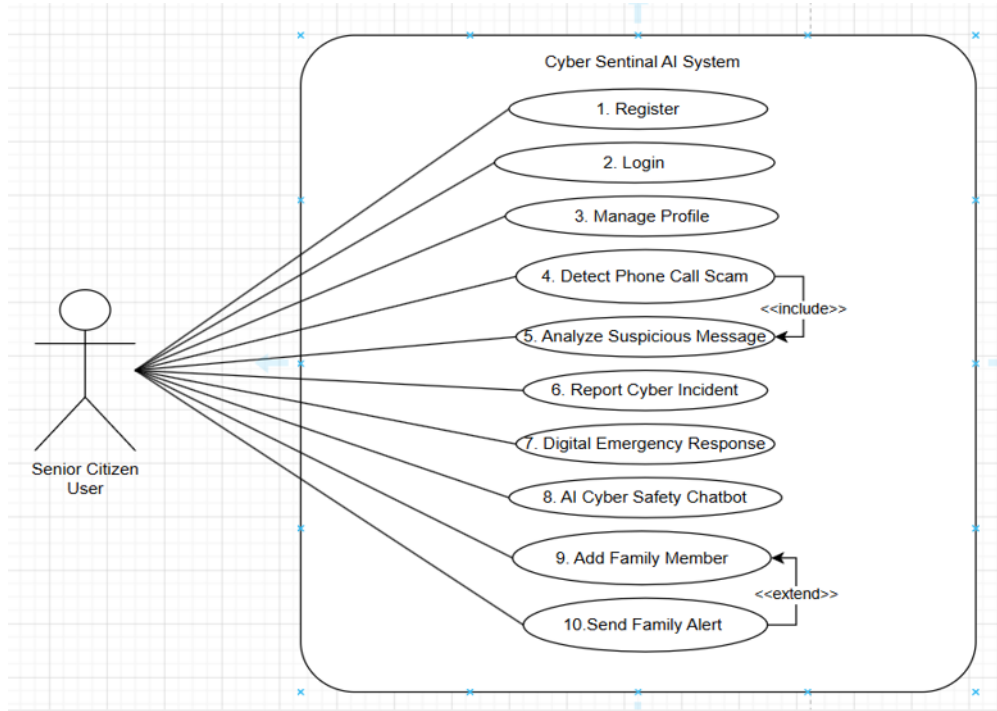
- Scam analysis result is displayed.

**Main Success Scenario**

1. Student uploads or enters job details.
2. System analyzes the recruitment information.
3. AI compares it with scam patterns.
4. Risk level is calculated.
5. Student receives recommendations.
6. Scam report is displayed.

**Exception Scenario**

- If job details are incomplete, analysis cannot be performed.
- If the AI service is unavailable, the system requests the user to try later.

**3. SENIOR CITIZEN USER****Use Case Description****Actor:** Senior Citizen User**Use Case:** Detect Phone Call Scam**Primary Actor:** Senior Citizen User**Secondary Actor:** AI Scam Detection System**Precondition:**

- User must be logged into the system.

**Postcondition:**

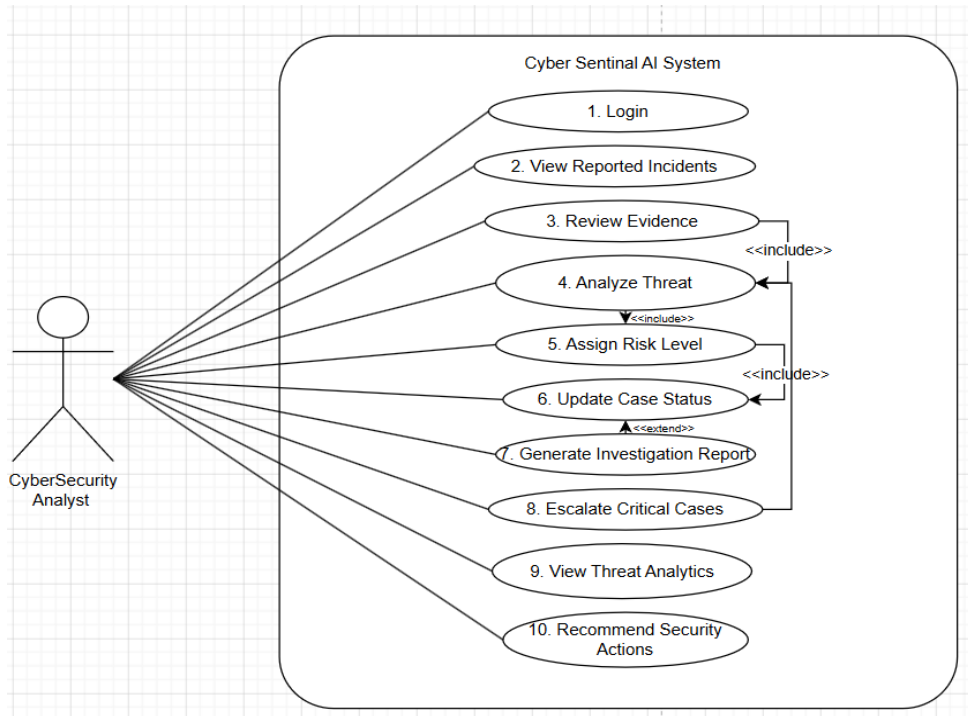
- Phone call risk level is displayed.

**Main Success Scenario**

1. User enters phone number or call details.
2. System analyzes the information.
3. AI detects possible scam patterns.
4. Risk level is generated.
5. User receives safety recommendations.
6. If required, a family alert is sent.

**Exception Scenario**

- Invalid call details are entered.
- AI service is temporarily unavailable.

**4.CYBERSECURITY ANALYST:****Use Case Description****Actor:** Cybersecurity Analyst**Use Case:** Review Reported Incidents**Primary Actor:** Cybersecurity Analyst**Secondary Actor:** Incident Management System**Precondition:**

- Analyst must log into the system.

**Postcondition:**

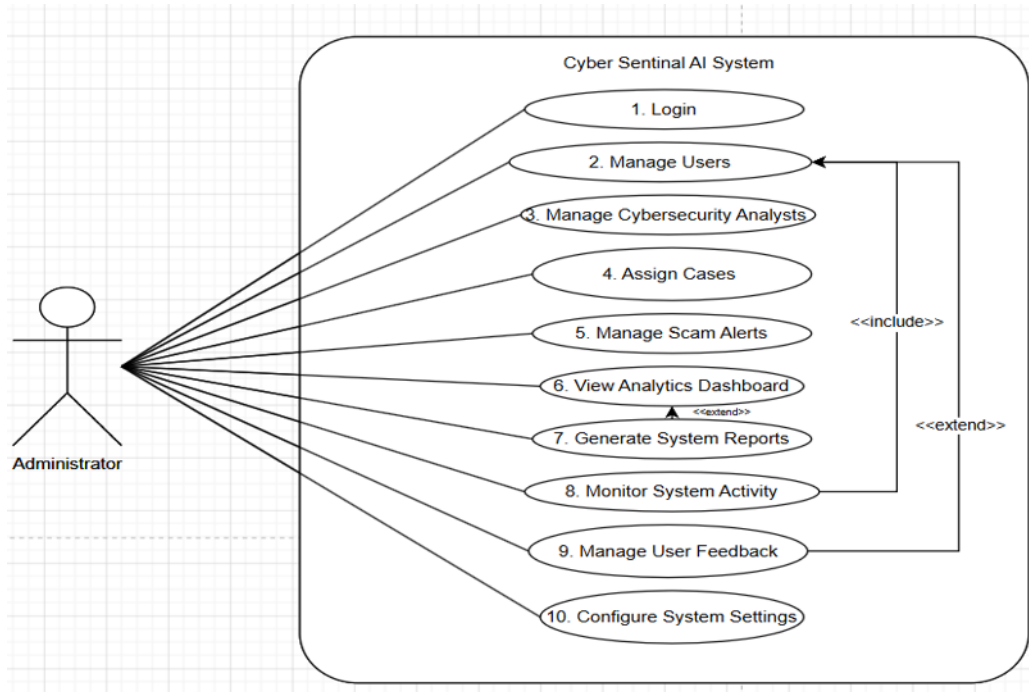
- Incident is reviewed and updated.

**Main Success Scenario**

1. Analyst logs into the system.
2. Opens the reported incidents list.
3. Reviews submitted evidence.
4. Analyzes the reported threat.
5. Updates the case status.
6. Saves the investigation report.

**Exception Scenario**

- Incident information is incomplete.
- Evidence is insufficient for analysis.

**5.ADMINISTRATOR:****Use Case Description****Actor:** Administrator**Use Case:** Manage Users**Primary Actor:** Administrator**Secondary Actor:** User Management System**Precondition:**

- Administrator must log into the system.

**Postcondition:**

- User information is successfully updated.

**Main Success Scenario**

1. Administrator logs into the system.
2. Opens the user management page.
3. Views registered users.
4. Adds, updates, or removes user accounts.
5. Saves the changes.
6. System confirms successful update.

### **Exception Scenario**

- Invalid user information is entered.
- User record cannot be found.
- Database connection fails.

## **2.4 Operating Environment**

*The Cyber Sentinel AI System operates in a web-based environment and can be accessed through modern web browsers.*

### ***Hardware Requirements***

- *Processor: Intel Core i3 or above*
- *RAM: Minimum 4 GB*
- *Storage: Minimum 10 GB free space*
- *Stable Internet Connection*

### ***Software Requirements***

#### ***Operating Systems:***

- *Windows 10/11*
- *Linux*
- *macOS*

#### ***Web Browsers:***

- *Google Chrome*
- *Mozilla Firefox*
- *Microsoft Edge*
- *Safari*

#### ***Server Environment:***

- *Node.js Runtime Environment*
- *Database Server (MySQL or MongoDB)*

### ***Development Environment***

- Visual Studio Code
- Git and GitHub
- Web Hosting Platform (Vercel or similar cloud service)

## **2.5 Design and Implementation Constraints**

*The development of the Cyber Sentinel AI System is subject to the following constraints:*

- *Compliance with cybersecurity and data privacy regulations.*
- *Use of secure authentication and authorization mechanisms.*
- *Protection of user data and uploaded evidence through encryption.*
- *Dependence on internet connectivity for real-time threat analysis.*
- *Compatibility with major web browsers and operating systems.*
- *Limited computational resources for AI model execution in the initial version.*
- *Use of approved development technologies such as HTML, CSS, JavaScript, React/Next.js, Node.js, and MySQL/MongoDB.*
- *Scalability requirements to support increasing numbers of users and reported incidents.*
- *Maintenance of secure communication using HTTPS protocols.*

## **2.6 User Documentation**

*The following documentation will be provided with the system:*

- *User Guide*
- *Administrator Manual*
- *Installation and Deployment Guide*

- *Online Help Documentation*
- *Frequently Asked Questions (FAQ)*
- *System Maintenance Manual*
- *API Documentation*
- *Cybersecurity Awareness Tutorials*
- *Troubleshooting Guide*

*All documentation will be provided in digital PDF format and accessible through the system help portal.*

## **2.7 Assumptions and Dependencies**

*The following assumptions and dependencies are considered during system development:*

### ***Assumptions***

- *Users have access to internet-enabled devices.*
- *Users possess basic knowledge of web applications.*
- *Threat analysis services and AI components remain available during operation.*
- *Users provide accurate information when reporting incidents.*
- *System administrators regularly monitor and maintain the platform.*

### ***Dependencies***

- *Dependence on web hosting infrastructure and cloud services.*
- *Dependence on database management systems for data storage.*
- *Dependence on AI and machine learning libraries for threat detection.*
- *Dependence on external cybersecurity threat intelligence sources for updated threat information.*

- *Dependence on secure network communication protocols.*
- *Dependence on third-party authentication and notification services if integrated.*

*Any significant changes in these assumptions or dependencies may affect system performance, functionality, or future development requirements.*

## **3. External Interface Requirements**

### **3.1 User Interfaces**

The Cyber Sentinel AI System shall provide a responsive, user-friendly, and role-based web interface for Registered Users, Student Users, Senior Citizen Users, Cybersecurity Analysts, and Administrators. After successful authentication, each user shall be directed to a dashboard based on their assigned role and shall have access only to the features permitted for that role. The common interface shall include login and registration pages, dashboards, navigation menus, profile management, notifications, data-entry forms, search and filter options, risk assessment displays, status indicators, loading indicators, confirmation dialogs, success messages, error messages, and logout functionality.

The Registered User interface shall support URL threat analysis, risk assessment, security recommendations, analysis history, and notifications, while the Student User interface shall support fake job and internship scam detection, recruitment message verification, scam reports, analysis history, and recommendations. The Senior Citizen interface shall provide phone-call scam detection, safety recommendations, emergency contact management, family alerts, and analysis history. The Cybersecurity Analyst interface shall provide reported incident review, evidence viewing, threat analysis, risk assignment, investigation notes, and case management. The Administrator interface shall provide user management, incident management, case assignment, system monitoring, and audit-log management. The system shall display appropriate validation, error, success, and authorization messages for invalid inputs, failed operations, and unauthorized access

### **3.2 Hardware Interfaces**

The Cyber Sentinel AI System is a web-based application and does not require any specialized or dedicated hardware. The system can be accessed using commonly available desktop computers, laptops, tablets, and smartphones through a modern web browser. The client device requires a processor, memory, display, keyboard or touchscreen, and network connectivity sufficient to access the application. The application server or cloud environment provides the required computing and storage resources for application processing, database operations, AI-based threat analysis, incident management, and user authentication.

### **3.3 Software Interfaces**

The Cyber Sentinel AI System shall interact with several software components required for its operation. The frontend shall be developed using a modern web framework such as Next.js with React and TypeScript, while Tailwind CSS may be used for responsive user interface design. Node.js shall provide the application runtime environment, and Prisma ORM shall provide the interface between the application services and the database. The system shall use a relational database such as PostgreSQL for production data storage, while SQLite may be used during local development where applicable. The database shall maintain user accounts, roles, URL analyses, scam reports, phone-call analyses, risk assessments, notifications, incidents, evidence, cases, investigation notes, and audit information. The AIThreatAnalysisService shall provide a reusable software interface for URL threat analysis, fake job and internship scam detection, recruitment message verification, and phone-call scam analysis. The system shall exchange structured input and output data between the user interface, backend APIs, AI analysis service, and database. Git and GitHub shall be used for source-code management, while a cloud deployment platform such as Vercel may be used for hosting the web application. The application shall maintain separation between the presentation, service, and database layers so that individual components can be modified or replaced without affecting the complete system.

### **3.4 Communications Interfaces**

The Cyber Sentinel AI System shall use standard web-based communication mechanisms for interaction between client devices, application services, databases, and external services. HTTPS shall be used for secure communication between the user's web browser and the deployed application. Frontend components shall communicate with backend services through HTTP/HTTPS-based API requests using structured formats such as JSON. Authentication requests, URL analysis requests, scam detection requests, incident submissions, notifications, case updates, and administrative operations shall be transmitted through protected API endpoints. Database communication shall be handled through the configured database connection and Prisma ORM. All sensitive information, including authentication credentials and user-related data, shall be transmitted through secure connections. The system shall validate incoming requests and reject malformed, unauthorized, or invalid requests. The communication mechanism shall support reliable request and response handling, appropriate error responses, and synchronization between the user interface and backend data.

## **4. System Features**

The Cyber Sentinel AI System provides a set of integrated cybersecurity features designed to detect potential threats, assess risks, provide security recommendations, and support incident management. The major features include user authentication and role management, URL threat detection, job and internship scam detection, recruitment message verification, phone-call scam detection, family alerts, cybersecurity incident reporting, analyst case management, administrator management, notifications, analysis history, and AI-based threat analysis. Each feature is designed according to the user's role and works together to provide a complete cybersecurity protection and incident management platform.

## 4.1 System Feature 1

### 4.1 User Authentication and Role Management

#### 4.1.1 Description and Priority

The User Authentication and Role Management feature provides secure registration, login, logout, profile management, and role-based access control for the Cyber Sentinel AI System. The feature is assigned High priority because authentication is required to protect user information and system functionality. The system shall support five primary roles: Registered User, Student User, Senior Citizen User, Cybersecurity Analyst, and Administrator.

#### 4.1.2 Stimulus/Response Sequences

When a user enters valid registration information, the system shall validate the submitted details, create the user account, and allow the user to authenticate. When a registered user enters valid login credentials, the system shall authenticate the user, identify the assigned role, and redirect the user to the appropriate dashboard. When invalid credentials are entered, the system shall reject the login attempt and display an appropriate error message. When an authenticated user logs out, the system shall terminate the active session and prevent further access to protected pages until the user authenticates again.

#### 4.1.3 Functional Requirements

- REQ-AUTH-01:** The system shall allow new users to register using valid personal
- REQ-AUTH-02:** The system shall authenticate users using valid credentials.
- REQ-AUTH-03:** The system shall assign and recognize the appropriate user role.
- REQ-AUTH-04:** The system shall provide role-based access to authorized features.
- REQ-AUTH-05:** The system shall prevent unauthorized users
- REQ-AUTH-06:** The system shall provide logout functionality.
- REQ-AUTH-07:** The system shall allow authenticated users to manage their profile
- REQ-AUTH-08:** The system shall display an appropriate error message when invalid

## 4.2 System Feature 2

### 4.2.1 Registered User – URL Threat Detection

#### 4.2.1.1 Description and Priority

The URL Threat Detection feature allows Registered Users to analyze website URLs and identify potential cybersecurity threats such as malicious URLs and phishing websites. The system shall validate the entered URL, perform AI-based threat analysis, generate a risk assessment, and provide appropriate security recommendations to the user. This feature is assigned **High priority** because it provides a core cybersecurity protection capability of the Cyber Sentinel AI System.

#### 4.2.1.2 Stimulus/Response Sequences

When a Registered User enters a website URL, the system shall validate the URL format and submit it for AI-based analysis. The AI Threat Analysis Service shall analyze the URL for malicious and phishing indicators and determine the associated risk level. The system shall generate a risk assessment and display the analysis result along with suitable security recommendations. If the URL is invalid or incomplete, the system shall display an appropriate validation error. If the analysis service is unavailable, the system shall notify the user and allow the analysis to be attempted again.

#### 4.2.2 Functional Requirements

- **REQ-URL-01:** The system shall allow authenticated Registered Users to enter a website URL for analysis.
- **REQ-URL-02:** The system shall validate the submitted URL.
- **REQ-URL-03:** The system shall analyze the URL using the AI Threat Analysis Service.
- **REQ-URL-04:** The system shall detect potential malicious and phishing indicators.
- **REQ-URL-05:** The system shall generate and display a risk assessment for the analyzed URL.

## 5. Other Nonfunctional Requirements

### 5.1 Performance Requirements

The Cyber Sentinel AI System shall provide responsive performance under normal operating conditions. Major application pages such as login, registration, dashboards, history, and reporting interfaces should load within approximately three seconds under a normal network connection. Standard user operations such as form submission, authentication, retrieving history, and loading notifications should normally receive a response within three seconds, excluding processing time associated with external AI or third-party services. AI-based analysis shall return results as soon as the configured analysis service completes processing. Database queries shall be optimized to avoid unnecessary delays, and the application shall minimize unnecessary API requests and repeated database operations. The system shall support the expected number of concurrent users for the intended academic and demonstration environment without significant performance degradation.

### 5.2 .Safety Requirements

The system shall provide clear and understandable security recommendations whenever a potential threat or scam is detected. Risk assessments shall be presented as advisory information and shall not be represented as an absolute guarantee that a website, message, phone call, or other activity is completely safe or malicious. High and critical risk results shall be clearly communicated to users so that appropriate precautions can be taken. The system shall prevent unauthorized modification or deletion of important incident and case information. Sensitive information such as personal details, emergency contacts, evidence, and investigation

information shall not be unnecessarily exposed through the interface or error messages. The system shall provide appropriate safeguards for actions that may affect user accounts, incidents, cases, and family alerts.

### **5.3 Security Requirement**

The Cyber Sentinel AI System shall implement secure authentication and role-based authorization to protect application functionality and user information. Passwords shall never be stored in plain text and authentication credentials shall be handled using appropriate security mechanisms. Protected frontend routes and backend API endpoints shall verify authentication and authorization before allowing access to sensitive resources. Registered Users, Student Users, Senior Citizen Users, Cybersecurity Analysts, and Administrators shall only access the information and operations permitted for their roles. User-specific information such as analysis history, reports, notifications, and emergency contacts shall be isolated from other users. Input data shall be validated and sanitized before processing to reduce the risk of malicious input and common web attacks. Sensitive configuration values, database credentials, authentication secrets, and API keys shall be stored securely through environment variables or an appropriate secret-management mechanism. Production communication shall use HTTPS, and security-relevant administrative operations shall be recorded in audit logs.

### **5.4 Software Quality Attributes**

The Cyber Sentinel AI System shall emphasize usability, reliability, maintainability, scalability, security, portability, reusability, robustness, and testability. The interface shall be simple and understandable, particularly for Senior Citizen Users, while maintaining a professional appearance for analysts and administrators. The system shall handle invalid input, service failures, database errors, and unauthorized operations gracefully without causing application crashes or data loss. The software shall use modular architecture and clearly separated frontend, API, service, and database layers to improve maintainability. The AIThreatAnalysisService shall be reusable across multiple cybersecurity analysis features and shall allow future integration of more advanced AI or machine-learning models. The architecture shall be scalable enough to support additional threat detection capabilities and future system enhancements without requiring major changes to the existing modules. The application shall be compatible with commonly used modern web browsers and shall provide responsive behavior across desktop, tablet, and mobile devices. The system shall also be structured so that individual modules, services, APIs, and database operations can be independently tested.

### **5.5 Business Rules**

The Cyber Sentinel AI System shall operate according to defined role-based business rules. A user must authenticate before accessing protected functionality, and every authenticated account shall have an assigned role. Registered Users shall be permitted to perform URL threat analysis, view their analysis history, and receive security recommendations. Student Users shall be permitted to detect fake job and internship scams, verify recruitment messages, view scam reports, and access their analysis history. Senior Citizen Users shall be permitted to analyze suspicious phone calls, manage emergency or family contacts, receive safety recommendations, and use the family alert functionality. Family members shall not be treated as a separate system role. Cybersecurity Analysts shall be authorized to review reported incidents, examine evidence, analyze threats, manage cases, update case status, and close investigations. Administrators shall be authorized to manage users, roles, incidents, case assignments, system activity, and audit information. Users

shall not be permitted to access another user's private information, analysis history, notifications, reports, or emergency contact information. Only authorized analysts shall modify investigation and case information, while administrative functions shall be restricted to authorized Administrators.

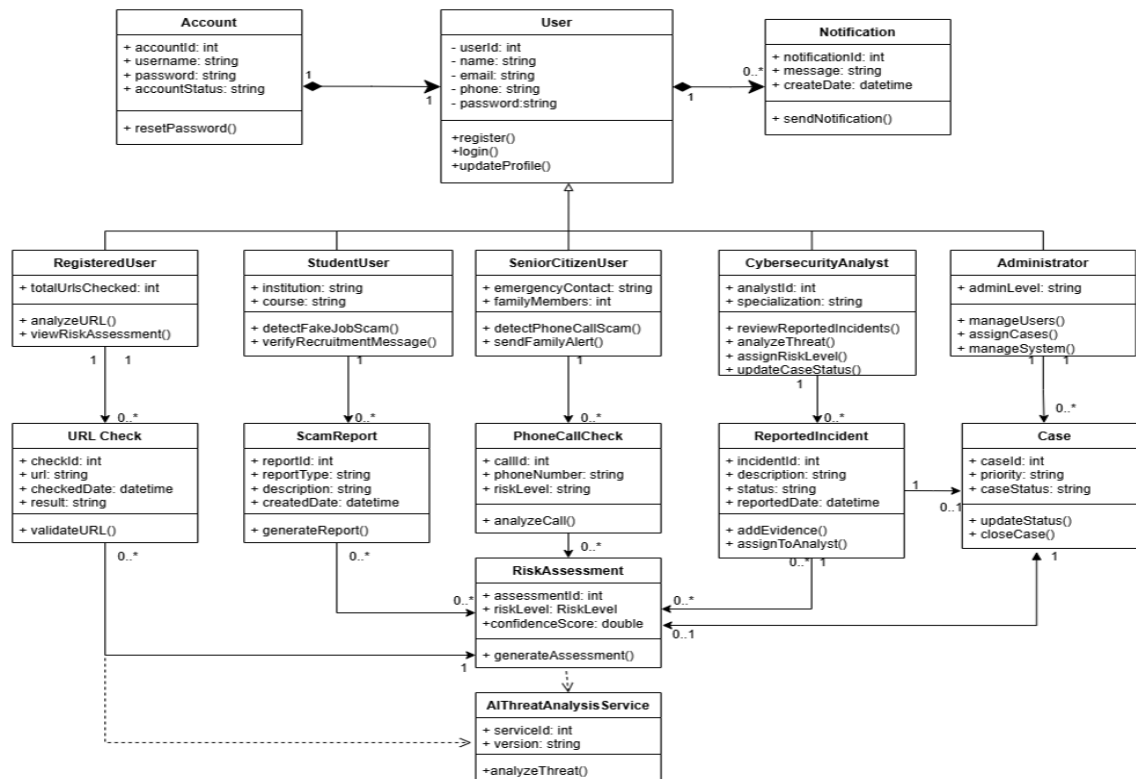
## 6. Other Requirements

The Cyber Sentinel AI System shall maintain a secure and reliable database for storing user information, authentication details, threat analysis results, incident reports, case records, notifications, and system logs. The system shall protect stored data from unauthorized access and shall support regular data backup and recovery. The application shall be designed for future extension to support additional user roles, cybersecurity features, AI analysis capabilities, and external security services without major changes to the existing system. The system shall use secure data-handling practices and comply with applicable privacy and security requirements. The interface shall be designed to support clear and understandable content, allowing future integration of additional languages if required.

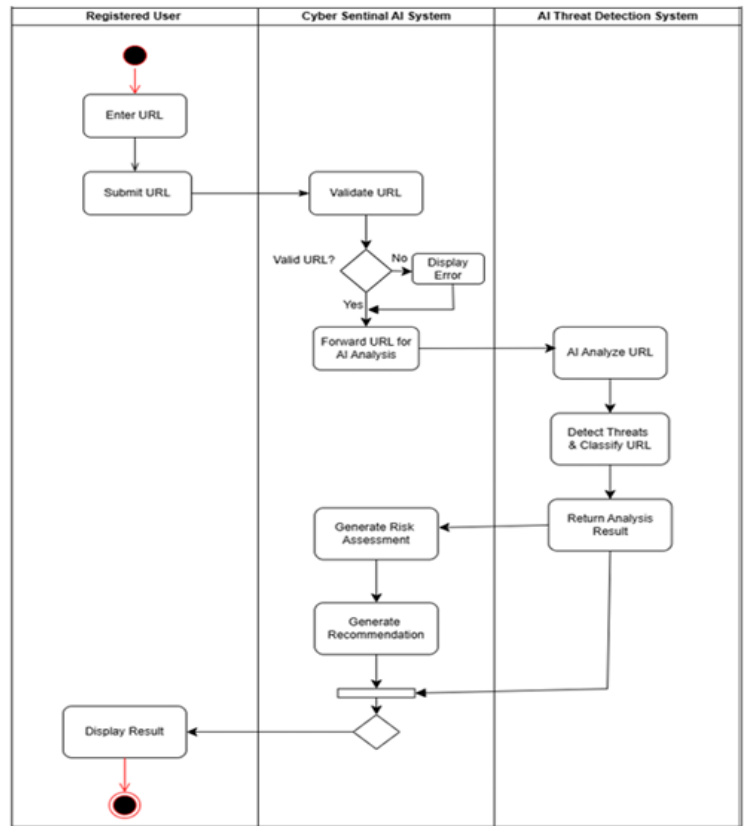
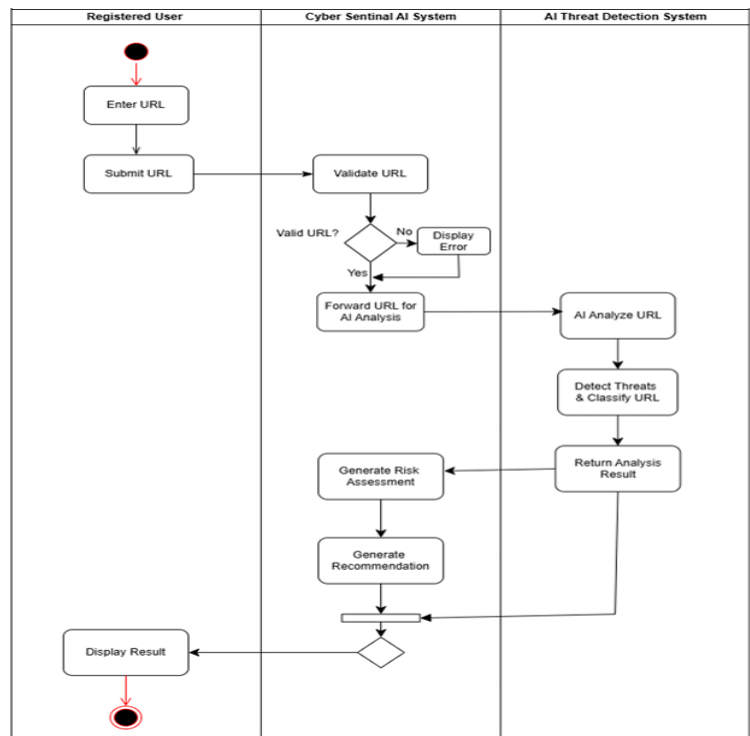
## Appendix B: Analysis Models

This appendix includes diagrams developed during system analysis and design

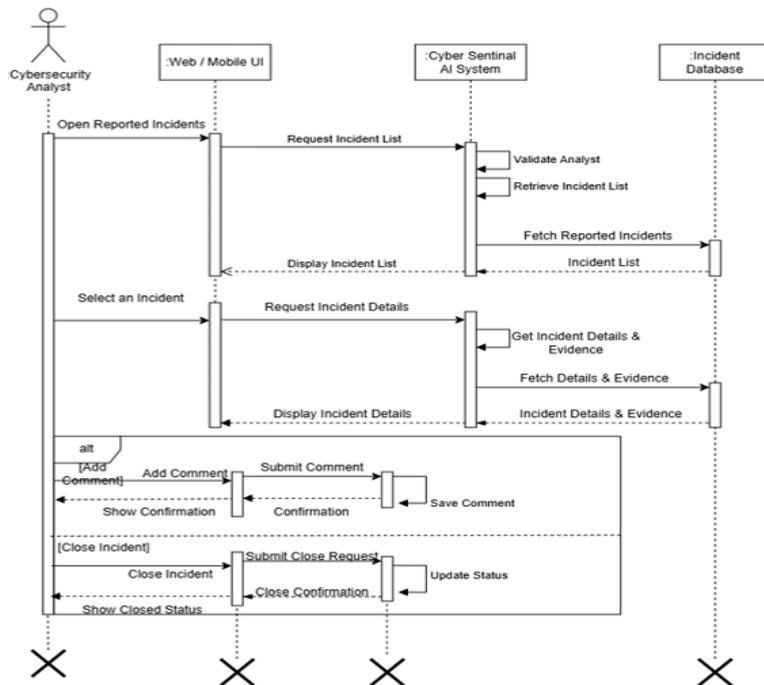
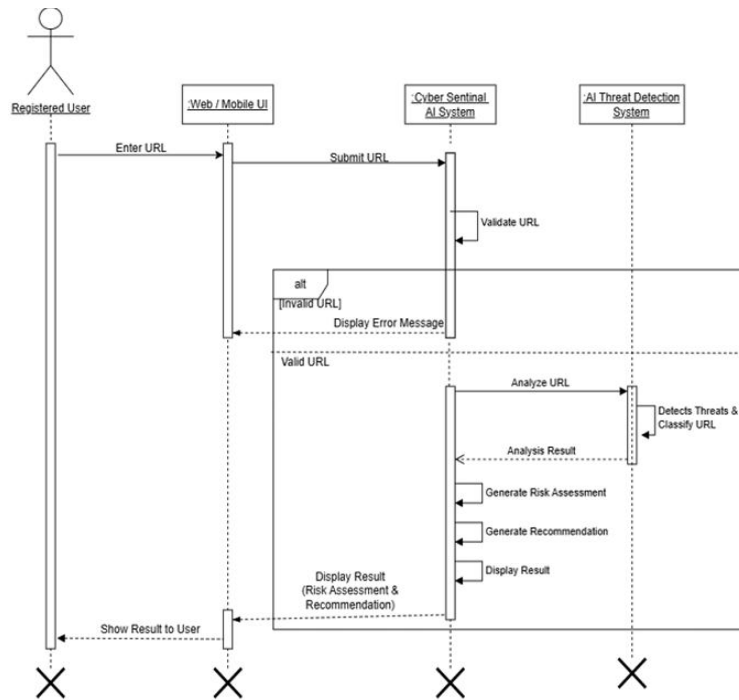
### 1. Class Diagram:



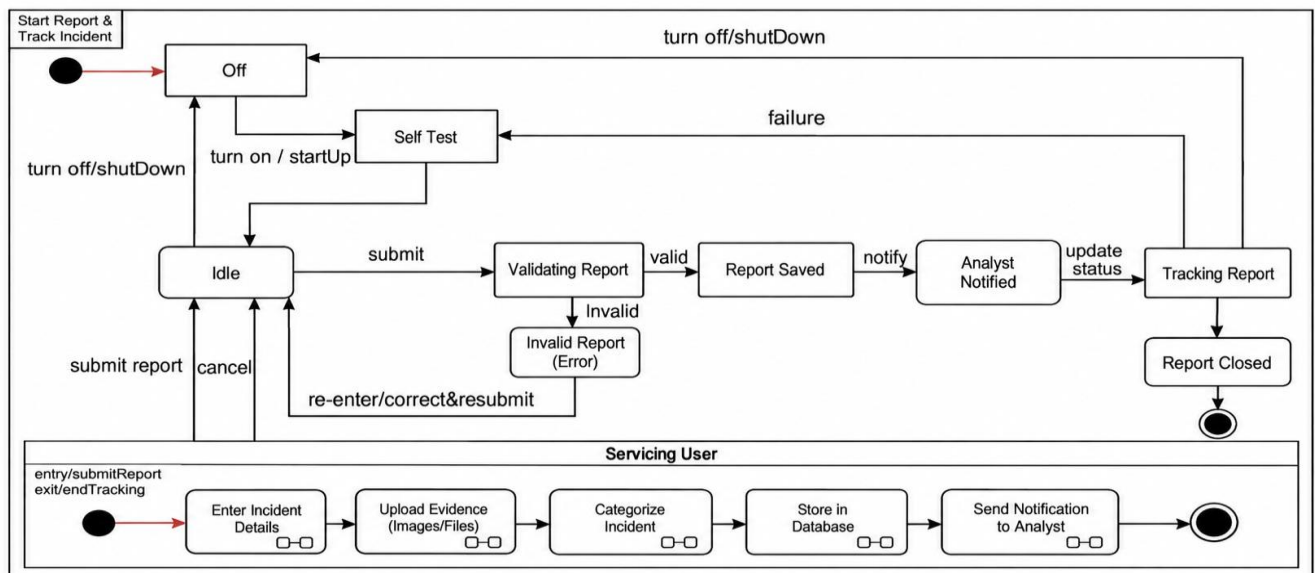
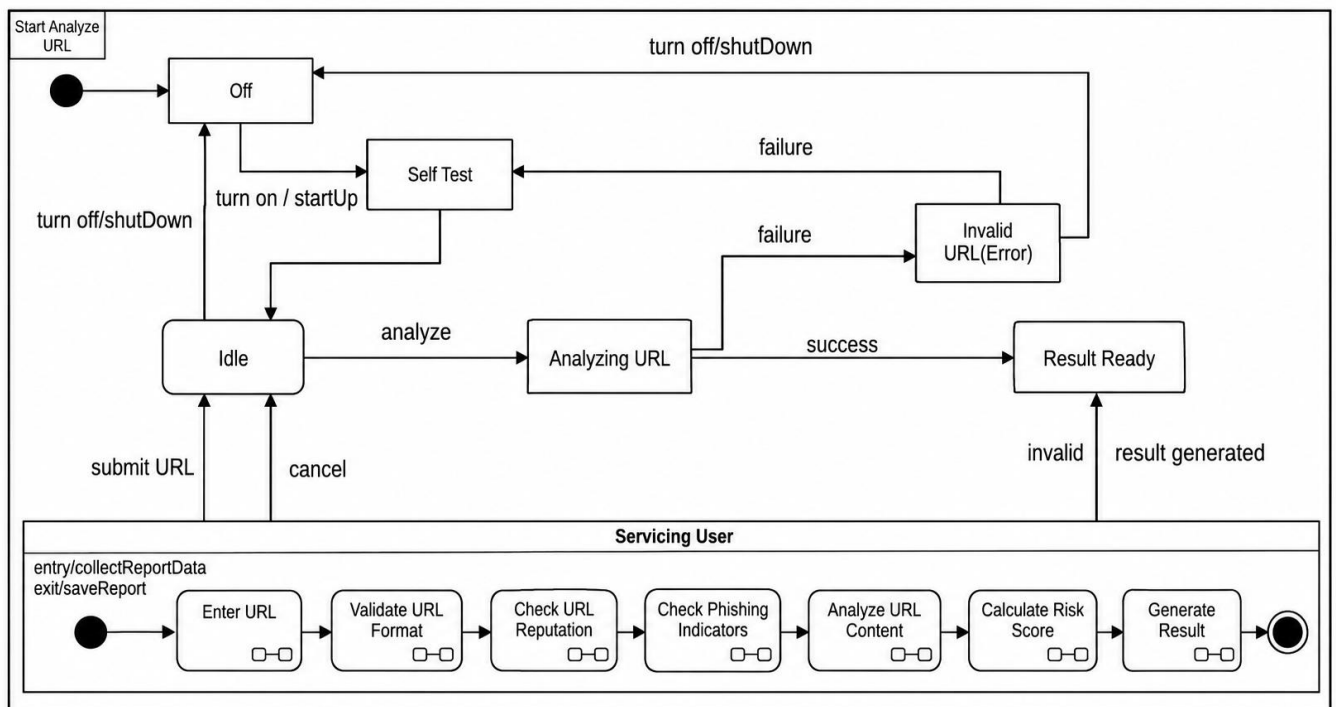
2.Activity Diagram:



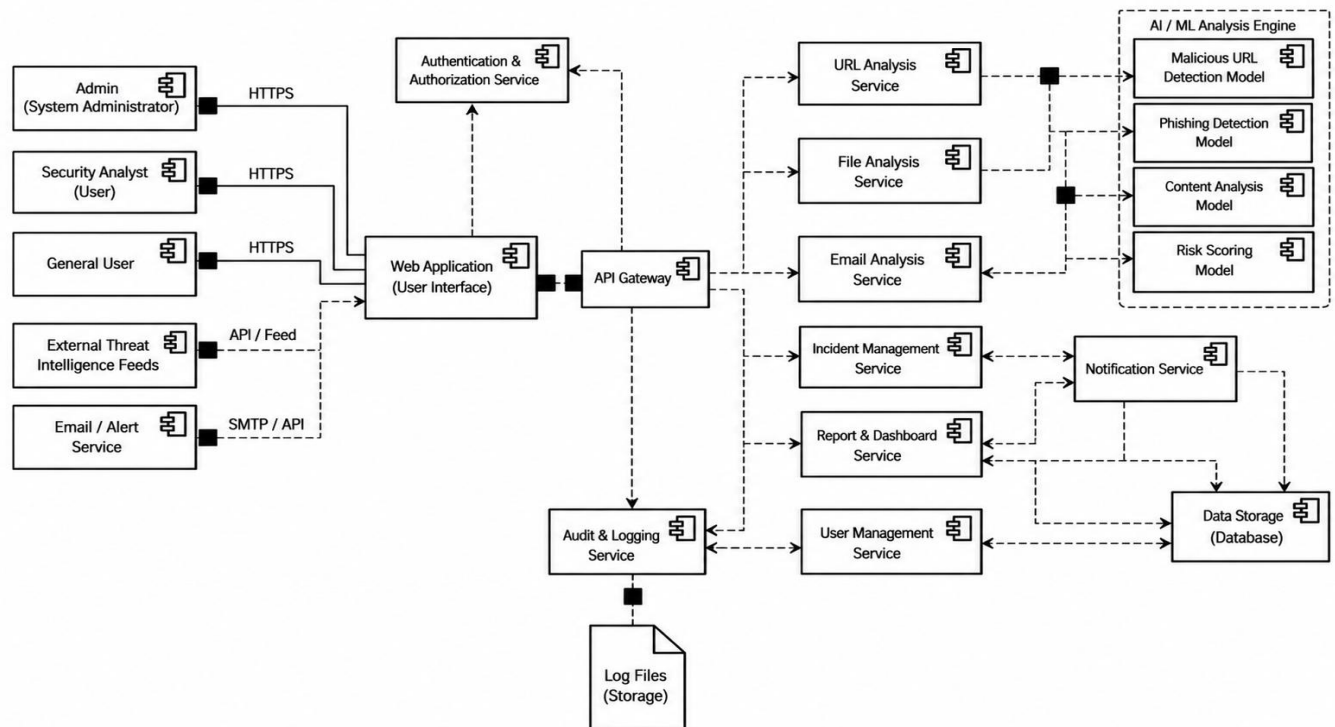
### 3. Sequence Diagram:



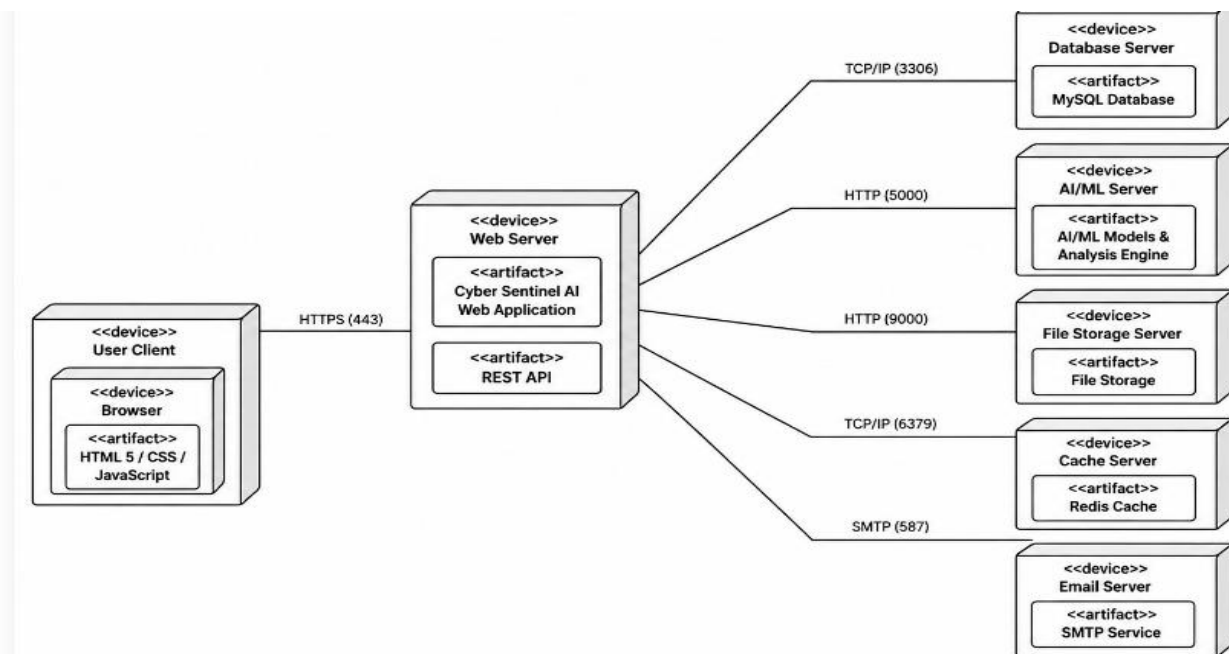
## 4. State Chart Diagram



## 5.Component Diagram:



## 6.Deployment Diagram:



**Source Code:****Schema.prisma:**

```

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}
generator client {
  provider = "prisma-client-js"
}
// User roles: "registered_user" | "student_user" |
// "senior_citizen" | "cybersecurity_analyst" |
// "administrator"
// Risk levels: "SAFE" | "LOW" | "MEDIUM" |
// "HIGH" | "CRITICAL"
// User status: "ACTIVE" | "INACTIVE" |
// "SUSPENDED"
model User {
  id          String      @id @default(uuid())
  name        String
  email       String      @unique
  phone       String?
  passwordHash String
  role        String      @default("registered user")
  status      String      @default("ACTIVE") //
  ACTIVE, INACTIVE, SUSPENDED
  createdAt   DateTime     @default(now())
  updatedAt   DateTime     @updatedAt
  account     Account?
  notifications Notification[]
  urlChecks   URLCheck[]
  scamReports ScamReport[]
  phoneCallChecks PhoneCallCheck[]
  emergencyContacts EmergencyContact[]
  reportedIncidents ReportedIncident[]
  @relation("ReportedByUser")
  assignedIncidents ReportedIncident[]
  @relation("AssignedAnalystIncident")
  assignedCases Case[]
  @relation("AssignedAnalystCase")
  investigationNotes InvestigationNote[]
  auditLogs   AuditLog[]
}
model Account {
  id          String      @id @default(uuid())
  userId      String      @unique
  bio         String?
  institution  String?
  department   String?
  notificationEmailOptIn Boolean @default(true)
  createdAt   DateTime     @default(now())
  updatedAt   DateTime     @updatedAt
  user User @relation(fields: [userId], references:
[id], onDelete: Cascade)
}
model EmergencyContact {
  id          String      @id @default(uuid())
  userId      String
  name        String
  phone       String

```

```

  email       String?
  relationship String      @default("CHILD") //
  CHILD, SPOUSE, CAREGIVER, FRIEND,
  OTHER
  notifyOnCritical Boolean @default(true)
  createdAt   DateTime     @default(now())
  updatedAt   DateTime     @updatedAt

  user User @relation(fields: [userId], references:
[id], onDelete: Cascade)
}
model Notification {
  id          String      @id @default(uuid())
  userId      String
  message     String
  type        String      @default("ALERT") //
  "ANALYSIS_COMPLETE",
  "HIGH RISK DETECTED",
  "SECURITY_ALERT",
  "RECOMMENDATION", "FAMILY_ALERT",
  "CASE_UPDATE"
  read        Boolean     @default(false)
  createdAt   DateTime     @default(now())
  user User @relation(fields: [userId], references:
[id], onDelete: Cascade)
}
model URLCheck {
  id          String      @id @default(uuid())
  userId      String
  url         String
  checkedDate DateTime     @default(now())
  riskLevel   String      @default("SAFE")
  confidenceScore Float     @default(0.0)
  threatType  String?
  indicators  String       // Stored as JSON
  string
  explanation String
  recommendation String
  user       User          @relation(fields:
[userId], references: [id], onDelete: Cascade)
  riskAssessments RiskAssessment[]
}
model ScamReport {
  id          String      @id @default(uuid())
  userId      String
  reportType  String      // "JOB_SCAM" |
  "RECRUITMENT_MESSAGE"
  companyName String?
  jobTitle    String?
  jobDescription String?
  recruiterName String?
  recruiterEmail String?
  contactNumber String?
  recruitmentMessage String?
  salaryOffer String?
  websiteUrl  String?
  additionalDetails String?
  riskLevel   String      @default("SAFE")
  confidenceScore Float     @default(0.0)

```

```

    scamIndicators String // Stored as JSON
  string
    explanation String
    recommendation String
    createdAt DateTime @default(now())
    user User @relation(fields:
[userId], references: [id], onDelete: Cascade)
    riskAssessments RiskAssessment[]
  }
  model PhoneCallCheck {
    id String @id @default(uuid())
    userId String
    callerNumber String
    callerName String?
    transcript String?
    riskLevel String @default("SAFE")
    confidenceScore Float @default(0.0)
    threatType String?
    indicators String // Stored as JSON
  string
    explanation String?
    recommendation String?
    familyAlertSent Boolean @default(false)
    createdAt DateTime @default(now())
    user User @relation(fields:
[userId], references: [id], onDelete: Cascade)
    riskAssessments RiskAssessment[]
  }
  model RiskAssessment {
    id String @id @default(uuid())
    urlCheckId String?
    scamReportId String?
    phoneCallCheckId String?
    riskLevel String
    confidenceScore Float
    threatType String?
    indicators String // Stored as JSON
  string
    explanation String
    recommendation String
    createdAt DateTime @default(now())
    urlCheck URLCheck? @relation(fields:
[urlCheckId], references: [id], onDelete: Cascade)
    scamReport ScamReport?
    @relation(fields: [scamReportId], references: [id],
onDelete: Cascade)
    phoneCallCheck PhoneCallCheck?
    @relation(fields: [phoneCallCheckId], references:
[id], onDelete: Cascade)
  }
  model ReportedIncident {
    id String @id @default(uuid())
    userId String
    title String
    description String
    incidentType String @default("OTHER") //
PHISHING, JOB SCAM, PHONE SCAM,
IDENTITY THEFT, MALWARE, OTHER
    evidence String? // Text, links, or JSON
    metadata
    riskLevel String @default("MEDIUM") //
LOW, MEDIUM, HIGH, CRITICAL

```

```

    status String @default("OPEN") // OPEN,
UNDER REVIEW, IN PROGRESS,
RESOLVED, CLOSED
    priority String @default("MEDIUM") //
LOW, MEDIUM, HIGH, CRITICAL
    assignedToId String?
    resolution String?
    createdAt DateTime @default(now())
    updatedAt DateTime @updatedAt
    reporter User @relation("ReportedByUser",
fields: [userId], references: [id], onDelete:
Cascade)
    assignedTo User?
    @relation("AssignedAnalystIncident", fields:
[assignedToId], references: [id])
    cases Case[]
    notes InvestigationNote[]
  }
  model Case {
    id String @id @default(uuid())
    incidentId String
    analystId String?
    title String?
    status String @default("OPEN") // OPEN,
UNDER REVIEW, IN PROGRESS,
RESOLVED, CLOSED
    priority String @default("MEDIUM") //
LOW, MEDIUM, HIGH, CRITICAL
    notes String? // Summary notes
    createdAt DateTime @default(now())
    updatedAt DateTime @updatedAt
    incident ReportedIncident @relation(fields:
[incidentId], references: [id], onDelete: Cascade)
    analyst User?
    @relation("AssignedAnalystCase", fields:
[analystId], references: [id])
    caseNotes InvestigationNote[]
  }
}

```

### index.ts:

```

export type UserRole =
  | 'registered_user'
  | 'student_user'
  | 'senior_citizen'
  | 'cybersecurity_analyst'
  | 'administrator';
export type UserStatus = 'ACTIVE' | 'INACTIVE'
| 'SUSPENDED';
export type RiskLevel = 'SAFE' | 'LOW' |
'MEDIUM' | 'HIGH' | 'CRITICAL';
export type IncidentStatus = 'OPEN' |
'UNDER REVIEW' | 'IN PROGRESS' |
'RESOLVED' | 'CLOSED';
export type IncidentPriority = 'LOW' | 'MEDIUM'
| 'HIGH' | 'CRITICAL';
export interface UserSession {
  userId: string;
  name: string;
  email: string;
  phone?: string | null;
  role: UserRole;
  status?: UserStatus;
}
export interface IndicatorItem {

```

```

type: string;
title: string;
description: string;
severity: 'low' | 'medium' | 'high' | 'critical' | 'info';
}
export interface AnalysisResult {
  riskLevel: RiskLevel;
  confidenceScore: number;
  threatType: string;
  indicators: IndicatorItem[];
  explanation: string;
  recommendation: string;
  engineType: 'AI LLM' |
'SECURITY HEURISTICS ENGINE';
  familyAlertRecommended?: boolean;
}

export interface JobScamInput {
  companyName?: string;
  jobTitle?: string;
  jobDescription?: string;
  recruiterName?: string;
  recruiterEmail?: string;
  contactNumber?: string;
  recruitmentMessage?: string;
  salaryOffer?: string;
  websiteUrl?: string;
  additionalDetails?: string;
}
export interface PhoneCallInput {
  callerNumber?: string;
  callerName?: string;
  transcript?: string;
  callDuration?: string;
  callContext?: string;
}
export interface NotificationItem {
  id: string;
  userId: string;
  message: string;
  type: string;
  read: boolean;
  createdAt: string;
}
export interface EmergencyContactItem {
  id: string;
  userId: string;
  name: string;
  phone: string;
  email?: string | null;
  relationship: string;
  notifyOnCritical: boolean;
  createdAt: string;
}
export interface InvestigationNoteItem {
  id: string;
  caseId?: string | null;
  incidentId?: string | null;
  analystId: string;
  analystName?: string;
  note: string;
  createdAt: string;
}
export interface CaseItem {

```

```

  id: string;
  incidentId: string;
  analystId?: string | null;
  analystName?: string | null;
  title?: string | null;
  status: IncidentStatus;
  priority: IncidentPriority;
  notes?: string | null;
  createdAt: string;
  updatedAt: string;
  caseNotes?: InvestigationNoteItem[];
}
export interface ReportedIncidentItem {
  id: string;
  userId: string;
  reporterName?: string;
  reporterEmail?: string;
  title: string;
  description: string;
  incidentType: string;
  evidence?: string | null;
  riskLevel: RiskLevel;
  status: IncidentStatus;
  priority: IncidentPriority;
  assignedToId?: string | null;
  assignedToName?: string | null;
  resolution?: string | null;
  createdAt: string;
  updatedAt: string;
  cases?: CaseItem[];
  notes?: InvestigationNoteItem[];
}
export interface AuditLogItem {
  id: string;
  userId?: string | null;
  userName?: string | null;
  role?: string | null;
  action: string;
  details?: string | null;
  ipAddress?: string | null;
  timestamp: string;
}
export interface AdminUserItem {
  id: string;
  name: string;
  email: string;
  phone?: string | null;
  role: UserRole;
  status: UserStatus;
  createdAt: string;
  updatedAt: string;
  _count?: {
    urlChecks: number;
    scamReports: number;
    phoneCallChecks: number;
    reportedIncidents: number;
    assignedCases: number;
  };
}
model InvestigationNote {
  id String @id @default(uuid())
  caseId String?
  incidentId String?
  analystId String

```

```

    note String
    createdAt DateTime @default(now())
    case Case? @relation(fields: [caseId],
references: [id], onDelete: Cascade)
    incident ReportedIncident? @relation(fields:
[incidentId], references: [id], onDelete: Cascade)
    analyst User @relation(fields:
[analystId], references: [id], onDelete: Cascade)
}
model AuditLog {
    id String @id @default(uuid())
}
aiThreatAnalysisService.ts
import { AnalysisResult, IndicatorItem,
JobScamInput, PhoneCallInput, RiskLevel } from
'@/types';

/**
 * AIThreatAnalysisService
 *
 * Central reusable cybersecurity threat analysis
service for Cyber Sentinel AI System.
 * Supports:
 * - URL threat analysis & phishing detection
 * - Fake job/internship scam detection
 * - Recruitment message verification
 * - Phone call scam & suspicious caller analysis
(Senior Citizen protection)
 *
 * Architecture features a modular pattern analysis
engine with optional external AI LLM integration
capability.
 */
export class AIThreatAnalysisService {
    private static version = '2.0.0';

    /**
     * Analyzes a URL for cybersecurity threats,
phishing, deceptive domains, and malware
indicators.
     */
    public static async analyzeUrl(inputUrl: string):
Promise<AnalysisResult> {
        const trimmedUrl = inputUrl ? inputUrl.trim() :
'';

        // URL Validation
        if (!trimmedUrl) {
            throw new Error('Please enter a valid website
URL.');
```

```

        }
        let parsedUrl: URL;
        try {
            const urlToParse =
/^https?:\W/i.test(trimmedUrl) ? trimmedUrl :
`http://${trimmedUrl}`;
            parsedUrl = new URL(urlToParse);
        } catch {
            throw new Error('Please enter a valid website
URL format (e.g., https://example.com).');
```

```

        }
        const protocol =
parsedUrl.protocol.toLowerCase();
        if (protocol !== 'http:' && protocol !== 'https:')

        throw new Error('Invalid URL protocol. Only
HTTP and HTTPS URLs are supported.');
```

```

        }
        const hostname =
parsedUrl.hostname.toLowerCase();
        const pathname =
parsedUrl.pathname.toLowerCase();
        const fullUrl = parsedUrl.href.toLowerCase();
        const indicators: IndicatorItem[] = [];
        let riskScore = 0; // 0 to 100
        // 1. Check for IP Address hostname
        const isIpAddress =
/^(d{1,3}\.){3}d{1,3}$/.test(hostname);
        if (isIpAddress) {
            riskScore += 35;
            indicators.push({
                type: 'IP_HOST_DETECTED',
                title: 'Raw IP Address Hostname',
                description: 'The URL uses a numerical IP
address instead of a standard registered domain
name. Legitimate services rarely request users to
log in via raw IP addresses.',
                severity: 'high',
            });
        }
        // 2. Check for suspicious TLDs
        const suspiciousTlds = ['.tk', '.ml', '.ga', '.cf',
'.gq', '.xyz', '.top', '.vip', '.work', '.click', '.cc',
'.support', '.buzz', '.monster', '.icu', '.fit', '.rest'];
        const hasSuspiciousTld =
suspiciousTlds.some((tld) =>
hostname.endsWith(tld));
        if (hasSuspiciousTld) {
            riskScore += 25;
            indicators.push({
                type: 'HIGH_RISK_TLD',
                title: 'High-Risk Top-Level Domain',
                description: 'The domain uses an extension
frequently associated with low-cost or disposable
phishing campaigns.',
                severity: 'medium',
            });
        }
        // 3. Check for Deceptive Brand Typosquatting
        / Target Brand Keywords
        const sensitiveBrandKeywords = [
            'paypal', 'bank', 'secure', 'login', 'verify',
            'account', 'update', 'signin', 'auth',
            'amazon', 'apple', 'microsoft', 'google', 'netflix',
            'chase', 'wellsfargo', 'support',
            'recover', 'wallet', 'crypto', 'binance', 'coinbase',
            'payment', 'payout'
        ]
    }
}

```

```

];
const matchedKeywords =
sensitiveBrandKeywords.filter((kw) =>
hostname.includes(kw));
if (matchedKeywords.length >= 2 ||
(matchedKeywords.length >= 1 &&
!hostname.endsWith('.com') &&
!hostname.endsWith('.org') &&
!hostname.endsWith('.gov') &&
!hostname.endsWith('.edu')) {
riskScore += 30;
indicators.push({
type: 'BRAND_TYPOSQUATTING',
title: 'Deceptive Brand & Security
Keywords',
description: 'URL contains sensitive
brand/security terms (${matchedKeywords.join(
')}) in an unverified domain structure, indicating
potential credential phishing.',
severity: 'high',
});
}
// 4. Excessive Subdomains
const domainParts = hostname.split('.');
if (domainParts.length > 3 && !isIpAddress) {
riskScore += 15;
indicators.push({
type: 'EXCESSIVE_SUBDOMAINS',
title: 'Deceptive Subdomain Depth',
description: 'The URL uses multiple layered
subdomains designed to obscure the actual root
target domain.',
severity: 'medium',
});
}
// 5. Insecure HTTP Protocol for Sensitive
Pages
if (protocol === 'http:' &&
(pathname.includes('login') ||
pathname.includes('auth') ||
pathname.includes('account') ||
matchedKeywords.length > 0)) {
riskScore += 20;
indicators.push({
type: 'INSECURE_PROTOCOL',
title: 'Unencrypted HTTP Connection',
description: 'The destination link uses
unencrypted HTTP without SSL encryption on
sensitive paths.',
severity: 'high',
});
}
// 6. URL Shortener Detection
const shortenerDomains = ['bit.ly', 'tinyurl.com',
't.co', 'is.gd', 'buff.ly', 'ow.ly', 'cutt.ly', 'rb.gy'];
if (shortenerDomains.some((sd) =>
hostname.includes(sd))) {
riskScore += 15;
indicators.push({
type:
'URL_SHORTENER_OBFUSCATION',
title: 'Shortened / Masked URL Link',

```

```

description: 'The link is obscured behind a
URL redirection service. The final destination
domain cannot be verified prior to clicking.',
severity: 'low',
});
}
// 7. Suspicious Path & Query Parameters
if (fullUrl.includes('@')) {
riskScore += 35;
indicators.push({
type: 'CREDENTIAL_EMBEDDED_URL',
title: 'Deceptive User-Auth "@" Syntax',
description: 'The URL contains an "@"
symbol. Modern web browsers interpret text
before "@" as authentication credentials,
disguising the true hostname.',
severity: 'critical',
});
}
}
if
(/^.(exe|scr|bat|cmd|vbs|ps1|apk|iso|zip)$/.test(path
name)) {
riskScore += 30;
indicators.push({
type:
'EXECUTABLE_DOWNLOAD_LINK',
title: 'Direct Executable/Archive Download
Link',
description: 'The path points directly to an
executable binary or compressed archive file
which could contain malware or ransomware
payload.',
severity: 'high',
});
}
// Determine Risk Level & Threat Classification
let riskLevel: RiskLevel = 'SAFE';
let threatType = 'BENIGN_URL';
if (riskScore >= 70) {
riskLevel = 'CRITICAL';
threatType =
'PHISHING_CREDENTIAL_HARVESTER';
} else if (riskScore >= 45) {
riskLevel = 'HIGH';
threatType =
'SUSPICIOUS_DECEPTIVE_SITE';
} else if (riskScore >= 25) {
riskLevel = 'MEDIUM';
threatType =
'POTENTIALLY_UNWANTED_REDIRECT';
} else if (riskScore > 0) {
riskLevel = 'LOW';
threatType = 'LOW_RISK_INDICATOR';
}
if (indicators.length === 0) {
indicators.push({
type: 'CLEAN_ANALYSIS',
title: 'Standard Security Profile',
description: 'No known phishing patterns,
deceptive subdomains, or dangerous executable
triggers detected.',
severity: 'info',
});
}

```

```

    }
    const confidenceScore = Math.min(99.0,
    Math.max(65.0, 75.0 + indicators.length * 5.0));
    let explanation = "";
    let recommendation = "";
    switch (riskLevel) {
    case 'CRITICAL':
        explanation = `CRITICAL THREAT
        DETECTED: The submitted URL (${hostname})
        exhibits multiple severe red flags including
        phishing keyword spoofing and deceptive
        structural syntax. High probability of malicious
        intent to steal credentials or personal data.`;
        recommendation = 'DO NOT CLICK OR
        VISIT THIS LINK. Do not enter any passwords,
        personal details, or financial credentials. Report
        this URL to your organization or IT
        administrator.';
        break;
    case 'HIGH':
        explanation = `HIGH RISK WARNING: The
        domain (${hostname}) shows strong
        characteristics of a deceptive site or malicious
        link. Contains unverified brand terms or
        suspicious top-level domain extensions.`;
        recommendation = 'Avoid interacting with
        this link. Verify the legitimate company URL by
        searching directly through an established search
        engine or official app.';
        break;
    case 'MEDIUM':
        explanation = `MODERATE RISK
        NOTICE: The URL includes minor suspicious
        elements such as redirection masking or deep
        subdomains. Proceed with caution.`;
        recommendation = 'Exercise caution before
        logging in or granting permissions on this site.
        Check for proper SSL certificates (https) and
        verify domain ownership.';
        break;
    case 'LOW':
        explanation = `LOW RISK NOTICE: Minor
        non-critical anomalies detected. No explicit
        malicious patterns confirmed.`;
        recommendation = 'Double check the domain
        name in your browser address bar before
        providing personal information.';
        break;
    case 'SAFE':
        default:
        explanation = `CLEAN SECURITY
        ASSESSMENT: The URL (${hostname}) has
        been scanned and displays legitimate structural
        characteristics with standard domain formatting.`;
        recommendation = 'The link appears safe
        based on structural and pattern analysis. Always
        maintain safe browsing habits.';
        break;
    }
    return {
        riskLevel,
        confidenceScore:
        Math.round(confidenceScore * 10) / 10,
        threatType,

```

```

        indicators,
        explanation,
        recommendation,
        engineType:
        'SECURITY_HEURISTICS_ENGINE',
    };
    }
    /**
     * Analyzes job/internship recruitment details for
     scam red flags, fraudulent recruiters, fee requests,
     and fake offers.
     */
    public static async analyzeJobScam(input:
    JobScamInput): Promise<AnalysisResult> {
        const company = (input.companyName ||
        "").trim();
        const title = (input.jobTitle || "").trim();
        const desc = (input.jobDescription || "").trim();
        const recruiterName = (input.recruiterName ||
        "").trim();
        const recruiterEmail = (input.recruiterEmail ||
        "").trim().toLowerCase();
        const phone = (input.contactNumber || "").trim();
        const message = (input.recruitmentMessage ||
        "").trim();
        const salary = (input.salaryOffer || "").trim();
        const website = (input.websiteUrl || "").trim();
        const details = (input.additionalDetails ||
        "").trim();
        const fullText = `${company} ${title} ${desc}
        ${recruiterName} ${recruiterEmail} ${phone}
        ${message} ${salary} ${website}
        ${details}`.toLowerCase();
        if (!fullText.trim()) {
            throw new Error('Please provide sufficient
            recruitment information for analysis.');
```

```

        }
        const indicators: IndicatorItem[] = [];
        let scamScore = 0;
        // 1. Upfront Payment / Security Deposit
        Request
        const paymentKeywords = [
            'security deposit', 'equipment fee', 'training
            fee', 'laptop fee', 'badge fee',
            'processing fee', 'wire transfer', 'crypto',
            'bitcoin', 'zelle', 'venmo', 'cashapp',
            'buy gift card', 'check deposit', 'reimburse
            equipment', 'pay for software'
        ];
        const matchedPayment =
        paymentKeywords.filter(kw =>
        fullText.includes(kw));
        if (matchedPayment.length > 0) {
            scamScore += 45;
            indicators.push({
                type: 'UPFRONT PAYMENT DEMAND',
                title: 'Upfront Fee / Payment Request
                Required',
                description: `Mentions payment or financial
                transaction keywords (${matchedPayment.join(
                ',')). Legitimate employers NEVER ask candidate
                job applicants to pay fees for background checks,
                equipment, or badges.`
            },
            {
                severity: 'critical',

```

```

    });
  }
  // 2. Free Public Email Domain for Corporate Recruiter
  const freeDomains = ['@gmail.com',
    '@yahoo.com', '@hotmail.com', '@outlook.com',
    '@aol.com', '@mail.com', '@proton.me',
    '@yandex.com'];
  if (recruiterEmail) {
    const isFreeEmail = freeDomains.some((fd) =>
      recruiterEmail.endsWith(fd));
    if (isFreeEmail && company && !['freelance',
      'private', 'self-employed'].some((kw) =>
        company.toLowerCase().includes(kw))) {
      scamScore += 30;
      indicators.push({
        type: 'FREE_EMAIL_RECRUITER',
        title: 'Public Email Used for Enterprise Hiring',
        description: `Recruiter email ($ {recruiterEmail}) uses a free public provider instead of an official corporate domain for company "$ {company}"`,
        severity: 'high',
      });
    }
  }
  // 3. Exclusive Telegram / WhatsApp / Signal Chat Interviewing
  const chatOnlyKeywords = ['telegram', 'whatsapp', 'signal', 'google hangouts', 'text-only interview', 'chat interview', 'no video interview'];
  const matchedChat = chatOnlyKeywords.filter((kw) =>
    fullText.includes(kw));
  if (matchedChat.length > 0) {
    scamScore += 25;
    indicators.push({
      type: 'CHAT_ONLY_INTERVIEW',
      title: 'Chat App / Messaging Only Interview',
      description: `Recruitment relies on informal chat apps ($ {matchedChat.join(', ')}) without formal video or in-person evaluation.`,
      severity: 'medium',
    });
  }
  // 4. Unrealistic High Salary / Instant Hiring Promises
  const salaryKeywords = ['100/hr', '150/hr', '200/hr', '$5000/week', '$10000/month', 'no experience required', 'instant job offer', 'hired immediately', 'work 1 hour a day'];
  const matchedSalary = salaryKeywords.filter((kw) =>
    fullText.includes(kw));
  if (matchedSalary.length > 0 || (salary && (salary.includes('1000') || salary.includes('5000')) && desc.toLowerCase().includes('entry level'))) {
    scamScore += 25;
    indicators.push({
      type: 'UNREALISTIC_COMPENSATION',
      title: 'Unrealistically High Pay for Minimal Work',

```

```

      description: 'Promises unusually high compensation with minimal required skills or immediate hiring without evaluation.',
      severity: 'high',
    });
  }
  // 5. Overly Urgent Wording
  const urgencyKeywords = ['urgent hiring', 'act immediately', 'limited spots', 'apply today or lose spot', 'immediate start mandatory'];
  if (urgencyKeywords.some((kw) =>
    fullText.includes(kw))) {
    scamScore += 15;
    indicators.push({
      type: 'ARTIFICIAL_URGENCY',
      title: 'High Pressure / Urgent Deadline Language',
      description: 'Uses high-pressure psychological tactics to force quick decisions before candidates can verify legitimacy.',
      severity: 'medium',
    });
  }
  // 6. Suspicious / Non-Existent Company Website
  if (website) {
    if (website.includes('bit.ly') || website.includes('tinyurl') || website.includes('.tk') || website.includes('.xyz')) {
      scamScore += 20;
      indicators.push({
        type: 'SUSPICIOUS_JOB_LINK',
        title: 'Suspicious / Masked Job Website Link',
        description: 'The job website link points to a high-risk TLD or shortened URL redirect.',
        severity: 'high',
      });
    }
  }
  // Determine Risk Level & Threat Classification
  let riskLevel: RiskLevel = 'SAFE';
  let threatType = 'LEGITIMATE_JOB_OFFER';
  if (scamScore >= 65) {
    riskLevel = 'CRITICAL';
    threatType = 'CONFIRMED_RECRUITMENT_SCAM';
  } else if (scamScore >= 40) {
    riskLevel = 'HIGH';
    threatType = 'HIGH_RISK_JOB_OFFER';
  } else if (scamScore >= 20) {
    riskLevel = 'MEDIUM';
    threatType = 'SUSPICIOUS_RECRUITMENT_INDICATORS';
  } else if (scamScore > 0) {
    riskLevel = 'LOW';
    threatType = 'LOW_RISK_NOTICE';
  }
  if (indicators.length === 0) {
    indicators.push({
      type: 'CLEAN_RECRUITMENT_PROFILE',
      title: 'Standard Job Profile',

```

```

        description: 'No known payment requests,
chat-only interview traps, or suspicious recruiter
domains detected.',
        severity: 'info',
    });
}
const confidenceScore = Math.min(98.0,
Math.max(70.0, 80.0 + indicators.length * 4.0));
let explanation = "";
let recommendation = "";
switch (riskLevel) {
    case 'CRITICAL':
        explanation = `HIGHLY LIKELY SCAM:
This job/internship listing contains severe red
flags, including requests for payment/deposit,
public recruiter emails, or chat-only interviews.`;
        recommendation = 'DO NOT SEND
MONEY, BANK DETAILS, OR COPIES OF
YOUR PASSPORT/ID. Cut off contact with the
recruiter immediately. Legitimate companies
never charge applicants fees.';
        break;
    case 'HIGH':
        explanation = `SUSPICIOUS
OPPORTUNITY: Several warning signs detected.
The recruiter contact details or offer promises
deviate significantly from standard hiring
practices.`;
        recommendation = 'Verify the employer
independently by visiting the official corporate
website careers page directly. Never conduct
interviews solely over text messaging apps.';
        break;
    case 'MEDIUM':
        explanation = `CAUTION ADVISED: Minor
recruitment anomalies detected. Ensure all terms
are documented in an official written offer letter
before proceeding.`;
        recommendation = 'Research the recruiter on
LinkedIn and confirm their official corporate
email address before sharing sensitive personal
details.';
        break;
    case 'LOW':
        explanation = `LOW SCAM
PROBABILITY: The listing appears standard,
though normal candidate due diligence is
recommended.`;
        recommendation = 'Ensure you communicate
through official company channels and review job
responsibilities carefully.';
        break;
    case 'SAFE':
    default:
        explanation = `LEGITIMATE PROFILE: No
fraudulent recruitment patterns detected in the
submitted job information.`;
        recommendation = 'Proceed with standard
application procedures. Remember to protect your
personal identity information until a formal
contract is signed.';
        break;
}
return {

```

```

        riskLevel,
        confidenceScore:
Math.round(confidenceScore * 10) / 10,
        threatType,
        indicators,
        explanation,
        recommendation,
        engineType:
'SECURITY_HEURISTICS_ENGINE',
    };
}

/**
 * Verifies raw recruitment messages (SMS,
WhatsApp, LinkedIn DM) for phishing, job
scams, and social engineering.
 */
public static async
analyzeRecruitmentMessage(message: string):
Promise<AnalysisResult> {
    if (!message || !message.trim()) {
        throw new Error('Please paste a recruitment
message to verify.');
```

```

    }
    return this.analyzeJobScam({
        recruitmentMessage: message,
        jobDescription: message,
    });
}

/**
 * Analyzes a suspicious phone call, voicemail,
or conversation transcript for senior fraud tactics.
 */
public static async analyzePhoneCall(input:
PhoneCallInput): Promise<AnalysisResult> {
    const callerNumber = (input.callerNumber ||
    "").trim();
    const callerName = (input.callerName ||
    "").trim();
    const transcript = (input.transcript || "").trim();
    const context = (input.callContext || "").trim();
    const fullText = `${callerNumber}
${callerName} ${transcript}
${context}`.toLowerCase();
    if (!callerNumber && !transcript && !context)
    {
        throw new Error('Please provide at least a
phone number or conversation details to
analyze.');
```

```

    }
    const indicators: IndicatorItem[] = [];
    let scamScore = 0;
    // 1. Government / Law Enforcement
    Impersonation & Arrest Threats
    const govKeywords = [
        'irs', 'internal revenue service', 'police
department', 'sheriff', 'arrest warrant',
        'fbi', 'customs and border', 'social security
administration', 'ssn suspended',
        'lawsuit filed', 'federal warrant', 'jail time',
        'arrest you today'
    ];
    const matchedGov = govKeywords.filter((kw)
=> fullText.includes(kw));

```

```

    if (matchedGov.length > 0) {
      scamScore += 45;
      indicators.push({
        type:
'GOVERNMENT_IMPERSONATION_THREAT
',
        title: 'Government / Law Enforcement
Impersonation',
        description: `Caller claims to represent
official authorities (${matchedGov.join(', ')}) with
threats of arrest or legal action. Official agencies
NEVER demand immediate payment over the
telephone.` ,
        severity: 'critical',
      });
    }
    // 2. Bank Fraud / Account Compromise
    Urgency
    const bankKeywords = [
      'bank fraud department', 'account frozen',
      'unauthorized charge', 'transfer to safe account',
      'compromised checking', 'verify your pin',
      'online banking password', 'move your funds'
    ];
    const matchedBank =
bankKeywords.filter((kw) =>
fullText.includes(kw));
    if (matchedBank.length > 0) {
      scamScore += 40;
      indicators.push({
        type:
'BANK_ACCOUNT TAKEOVER SCAM',
        title: 'Urgent Bank / Account Compromise
Claim',
        description: `Caller demands immediate
transfer to a "safe" account or asks for banking
PIN/passwords (${matchedBank.join(', ')}). Your
bank will NEVER ask you to transfer funds to
protect them.` ,
        severity: 'critical',
      });
    }
    // 3. Tech Support & Remote Desktop Software
    Traps
    const techKeywords = [
      'microsoft support', 'windows virus', 'computer
infected', 'anydesk', 'teamviewer',
      'quicksupport', 'ultraviewer', 'download
software to fix', 'allow remote access', 'call from
apple support'
    ];
    const matchedTech = techKeywords.filter((kw)
=> fullText.includes(kw));
    if (matchedTech.length > 0) {
      scamScore += 40;
      indicators.push({
        type:
'TECH SUPPORT REMOTE ACCESS',
        title: 'Fake Tech Support & Remote Control
Demand',
        description: `Caller instructs you to install
remote desktop software (${matchedTech.join(',
')}) to remove non-existent viruses. Granting

```

```

access allows attackers to take over your PC and
steal banking data.` ,
        severity: 'high',
      });
    }
    // 4. Grandchild / Family Emergency Impostor
    Scam
    const familyKeywords = [
      'grandma', 'grandpa', 'in jail', 'car accident',
      'need bail money', 'bonded out',
      'please don\'t tell mom', 'don\'t tell my parents',
      'lawyer needs cash'
    ];
    const matchedFamily =
familyKeywords.filter((kw) =>
fullText.includes(kw));
    if (matchedFamily.length > 0) {
      scamScore += 50;
      indicators.push({
        type:
'GRANDCHILD EMERGENCY IMPOSTOR',
        title: 'Grandparent / Family Emergency
Impersonation',
        description: `Uses emotional distress
claiming a relative is in jail or hospitalized and
asks you not to contact other family members.
Always verify directly with family before sending
funds.` ,
        severity: 'critical',
      });
    }
    // 5. Irreversible Payment Methods (Gift Cards,
    Crypto ATM, Wire)
    const paymentKeywords = [
      'gift card', 'target gift card', 'apple card',
      'google play card', 'walmart card',
      'bitcoin machine', 'crypto atm', 'wire transfer',
      'western union', 'moneygram', 'cash in envelope'
    ];
    const matchedPayment =
paymentKeywords.filter((kw) =>
fullText.includes(kw));
    if (matchedPayment.length > 0) {
      scamScore += 45;
      indicators.push({
        type:
'UNTRACEABLE PAYMENT DEMAND',
        title: 'Demanding Gift Cards or Crypto ATM
Payment',
        description: `Instructs you to buy gift cards
or visit a cryptocurrency kiosk
(${matchedPayment.join(', ')}). No legitimate
business, utility, or government agency accepts
gift cards as payment.` ,
        severity: 'critical',
      });
    }
    // 6. Extreme Psychological Pressure & Secrecy
    Tactics
    const pressureKeywords = [
      'do not hang up', 'stay on the phone', 'don\'t
talk to anyone', 'confidential order',
      'act immediately', 'within thirty minutes',
      'immediate arrest'
    ];

```

```

];
const matchedPressure =
pressureKeywords.filter((kw) =>
fullText.includes(kw));
if (matchedPressure.length > 0) {
  scamScore += 25;
  indicators.push({
    type: 'COERCIVE_SECRECY_TACTICS',
    title: 'High-Pressure Secrecy & Urgency
Tactics',
    description: 'Caller uses intimidation, forbids
you from hanging up, or demands you hide the
situation from family.',
    severity: 'high',
  });
}
// 7. Suspicious Caller Number Format / Known
Scam Patterns
if (callerNumber) {
  const cleanNum =
callerNumber.replace(/D/g, "");
  if (cleanNum.startsWith('800') ||
cleanNum.startsWith('888') ||
cleanNum.startsWith('877') ||
cleanNum.startsWith('866')) {
    scamScore += 10;
    indicators.push({
      type: 'TOLL_FREE_OUTBOUND_CALL',
      title: 'Toll-Free Outbound Caller ID',
      description: 'Incoming calls displaying toll-
free numbers are frequently spoofed by overseas
automated robocall centers.',
      severity: 'low',
    });
  }
}

// Determine Risk Level & Threat Classification
let riskLevel: RiskLevel = 'SAFE';
let threatType =
'STANDARD_PHONE_CALL';
if (scamScore >= 65) {
  riskLevel = 'CRITICAL';
  threatType =
'CONFIRMED_PHONE_FRAUD_SCAM';
} else if (scamScore >= 40) {
  riskLevel = 'HIGH';
  threatType =
'HIGH_RISK_SUSPICIOUS_CALL';
} else if (scamScore >= 20) {
  riskLevel = 'MEDIUM';
  threatType =
'POTENTIALLY_UNWANTED_ROBOCALL';
} else if (scamScore > 0) {
  riskLevel = 'LOW';
  threatType =
'LOW_RISK_CALL_INDICATOR';
}
if (indicators.length === 0) {
  indicators.push({
    type: 'CLEAN_CALL_PROFILE',
    title: 'No Immediate Scam Indicators',

```

```

    description: 'No known high-pressure arrest
threats, gift card demands, or remote desktop
software prompts detected.',
    severity: 'info',
  });
}
const confidenceScore = Math.min(99.0,
Math.max(70.0, 78.0 + indicators.length * 4.5));
let explanation = "";
let recommendation = "";
switch (riskLevel) {
  case 'CRITICAL':
    explanation = 'DANGEROUS SCAM CALL
DETECTED: This call exhibits severe indicators
of elder financial fraud, such as law enforcement
impersonation, grandchild emergency claims, or
gift card/crypto payment demands.';
    recommendation = 'HANG UP THE PHONE
IMMEDIATELY. Do not provide any bank
numbers, PINs, or gift card codes. Notify a trusted
family member or emergency contact right away.';
    break;
  case 'HIGH':
    explanation = 'HIGH RISK WARNING: The
caller is utilizing high-pressure urgency tactics,
requesting remote access to your computer, or
claiming an unverified account problem.';
    recommendation = 'Hang up and call the
official institution directly using the telephone
number on the back of your bank card or utility
statement.';
    break;
  case 'MEDIUM':
    explanation = 'SUSPICIOUS ACTIVITY:
This call contains unusual elements like toll-free
spoofing or unsolicited marketing urgency.';
    recommendation = 'Do not confirm your
name, address, or financial details. Ask for written
verification via official postal mail.';
    break;
  case 'LOW':
    explanation = 'LOW RISK NOTICE: Minor
non-critical indicators detected, but no aggressive
extortion patterns found.';
    recommendation = 'Never share two-factor
authentication codes or passwords over the
phone.';
    break;
  case 'SAFE':
  default:
    explanation = 'NORMAL CALL PROFILE:
The analyzed call details show standard
communication patterns without predatory scam
keywords.';
    recommendation = 'Stay vigilant. If a caller
ever asks for banking codes, gift cards, or money
transfers in the future, hang up immediately.';
    break;
}
return {
  riskLevel,
  confidenceScore:
Math.round(confidenceScore * 10) / 10,
  threatType,

```

```

    indicators,
    explanation,
    recommendation,
    engineType:
'SECURITY_HEURISTICS_ENGINE',
    familyAlertRecommended: riskLevel ===
'HIGH' || riskLevel === 'CRITICAL',
  };
}
}
middleware.ts:
import { NextResponse } from 'next/server';
import type { NextRequest } from 'next/server';
import { jwtVerify } from 'jose';
const JWT_SECRET_KEY = new
TextEncoder().encode(
  process.env.JWT_SECRET || 'cyber-sentinel-
super-secret-jwt-key-2026-secure-production-
hash'
);
function getRoleDefaultDashboard(role: string):
string {
  switch (role) {
    case 'student_user':
      return '/student/dashboard';
    case 'senior_citizen':
      return '/senior/dashboard';
    case 'cybersecurity_analyst':
      return '/analyst/dashboard';
    case 'administrator':
      return '/admin/dashboard';
    case 'registered_user':
    default:
      return '/registered/dashboard';
  }
}
export async function middleware(request:
NextRequest) {
  const { pathname } = request.nextUrl;
  const token =
request.cookies.get('cyber_sentinel_token')?.value
;
  let session: any = null;
  if (token) {
    try {
      const verified = await jwtVerify(token,
JWT_SECRET_KEY);
      session = verified.payload;
    } catch {
      session = null;
    }
  }
  const isAuthPage = pathname === '/login' ||
pathname === '/register';
  const isProtectedPath =
    pathname.startsWith('/registered') ||
    pathname.startsWith('/student') ||
    pathname.startsWith('/senior') ||
    pathname.startsWith('/analyst') ||
    pathname.startsWith('/admin') ||
    pathname.startsWith('/profile') ||
    pathname.startsWith('/report-incident');
  // 1. Unauthenticated user trying to access
protected routes

```

```

    if (isProtectedPath && !session) {
      const loginUrl = new URL('/login', request.url);
      loginUrl.searchParams.set('from', pathname);
      return NextResponse.redirect(loginUrl);
    }
    // 2. Authenticated user trying to access
login/register pages
    if (isAuthPage && session) {
      return NextResponse.redirect(new
URL(getRoleDefaultDashboard(session.role),
request.url));
    }
    // 3. Strict Role-Based Access Control (RBAC)
& Boundary Enforcement
    if (session) {
      // Administrator has access to all admin routes
& monitoring
      if (pathname.startsWith('/admin') &&
session.role !== 'administrator') {
        return NextResponse.redirect(new
URL(getRoleDefaultDashboard(session.role),
request.url));
      }
      // Cybersecurity Analyst routes
      if (pathname.startsWith('/analyst') &&
session.role !== 'cybersecurity_analyst' &&
session.role !== 'administrator') {
        return NextResponse.redirect(new
URL(getRoleDefaultDashboard(session.role),
request.url));
      }
      // Senior Citizen routes
      if (pathname.startsWith('/senior') &&
session.role !== 'senior_citizen' && session.role
!== 'administrator') {
        return NextResponse.redirect(new
URL(getRoleDefaultDashboard(session.role),
request.url));
      }
      // Student User routes
      if (pathname.startsWith('/student') &&
session.role !== 'student_user' && session.role
!== 'administrator') {
        return NextResponse.redirect(new
URL(getRoleDefaultDashboard(session.role),
request.url));
      }
      // Registered User routes
      if (pathname.startsWith('/registered') &&
session.role !== 'registered_user' && session.role
!== 'administrator') {
        return NextResponse.redirect(new
URL(getRoleDefaultDashboard(session.role),
request.url));
      }
    }
    return NextResponse.next();
  }
}
export const config = {
  matcher: [
    '/registered/:path*',
    '/student/:path*',
    '/senior/:path*',
    '/analyst/:path*',

```

```

'/admin/:path*',
'/profile/:path*',
'/report-incident/:path*',
'/login',
'/register',

```

### auth.ts:

```

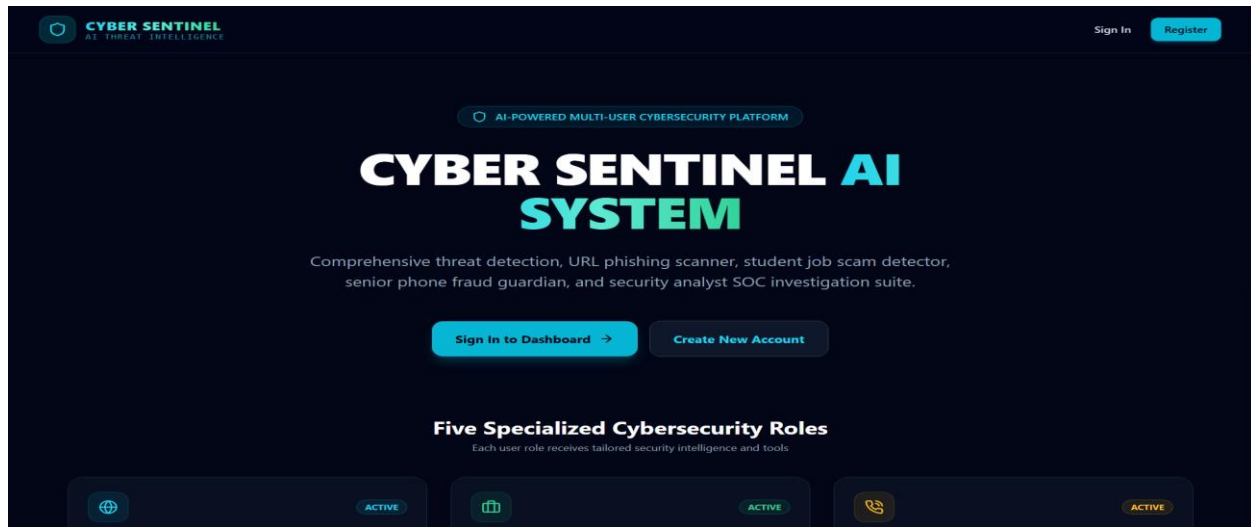
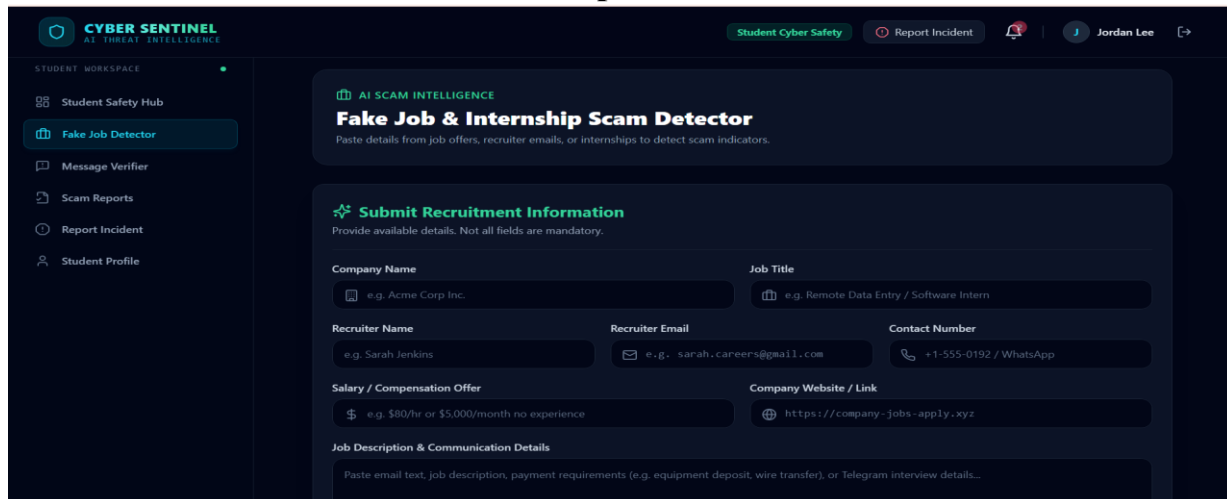
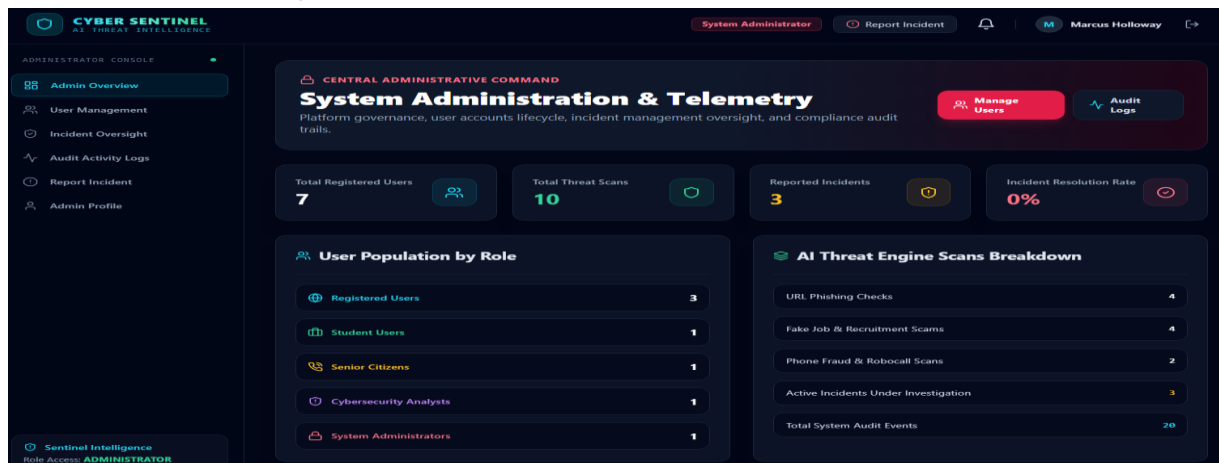
import { SignJWT, jwtVerify } from 'jose';
import bcrypt from 'bcryptjs';
import { cookies } from 'next/headers';
import { UserSession, UserRole, UserStatus }
from '@types';
import { db } from '@lib/db';
const JWT_SECRET_KEY = new
TextEncoder().encode(
  process.env.JWT_SECRET || 'cyber-sentinel-
super-secret-jwt-key-2026-secure-production-
hash'
);
const TOKEN_COOKIE_NAME =
'cyber_sentinel_token';
export async function hashPassword(password:
string): Promise<string> {
  const salt = await bcrypt.genSalt(10);
  return bcrypt.hash(password, salt);
}
export async function verifyPassword(password:
string, hash: string): Promise<boolean> {
  return bcrypt.compare(password, hash);
}
export async function signToken(payload:
UserSession): Promise<string> {
  return new SignJWT({ ...payload })
    .setProtectedHeader({ alg: 'HS256' })
    .setIssuedAt()
    .setExpirationTime('7d')
    .sign(JWT_SECRET_KEY);
}
export async function verifyToken(token: string):
Promise<UserSession | null> {
  try {
    const verified = await jwtVerify(token,
JWT_SECRET_KEY);
    return verified.payload as unknown as
UserSession;
  } catch (error) {
    return null;
  }
}
export async function setAuthCookie(session:
UserSession) {
  const token = await signToken(session);
  const cookieStore = cookies();
  cookieStore.set(TOKEN_COOKIE_NAME,
token, {

```

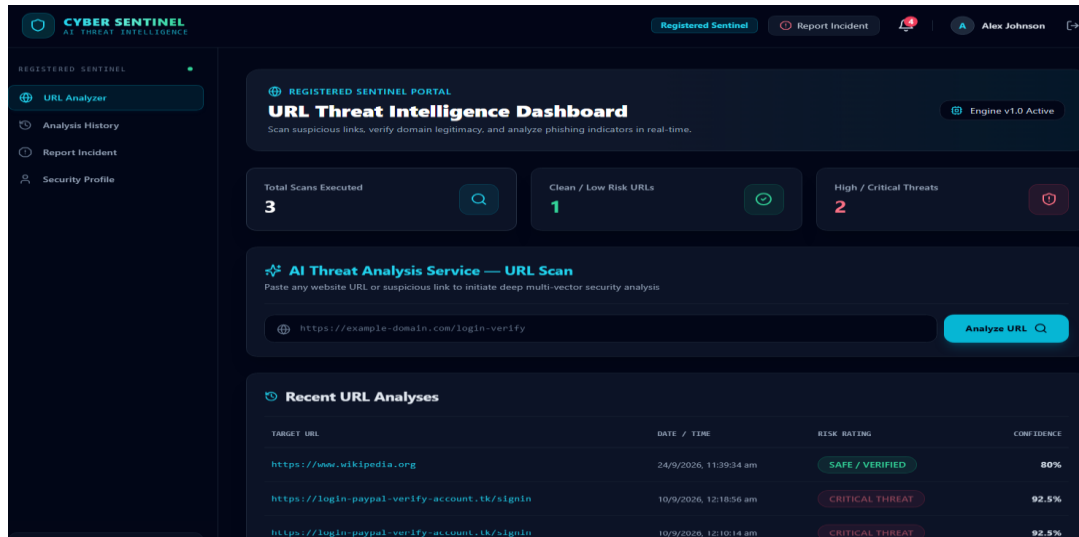
```

  },
  });
  httpOnly: true,
  secure: process.env.NODE_ENV ===
'production',
  sameSite: 'lax',
  path: '/',
  maxAge: 7 * 24 * 60 * 60, // 7 days
});
}
export async function clearAuthCookie() {
  const cookieStore = cookies();
  cookieStore.delete(TOKEN_COOKIE_NAME);
}
export async function getSession():
Promise<UserSession | null> {
  const cookieStore = cookies();
  const token =
cookieStore.get(TOKEN_COOKIE_NAME)?.val
ue;
  if (!token) return null;
  return verifyToken(token);
}
export async function getCurrentUser() {
  const session = await getSession();
  if (!session?.userId) return null;
  try {
    const user = await db.user.findUnique({
      where: { id: session.userId },
      select: {
        id: true,
        name: true,
        email: true,
        phone: true,
        role: true,
        status: true,
        createdAt: true,
        account: true,
      },
    });
    if (!user || user.status === 'INACTIVE' ||
user.status === 'SUSPENDED') {
      return null;
    }
    return user;
  } catch (error) {
    return null;
  }
}

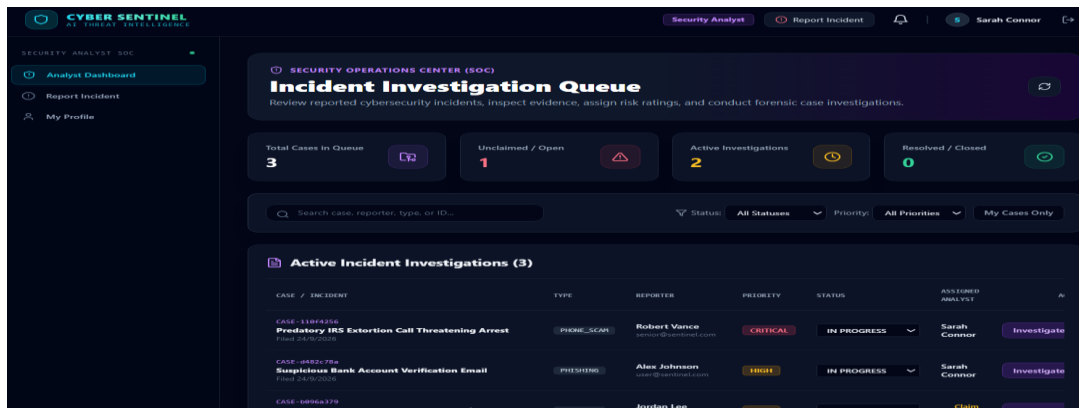
```

**DEMONSTRATION:****Cyber Sentinel AI System – Home Page****Student User – Fake Job & Internship Scam Detection****Administrator – System Administration Dashboard**

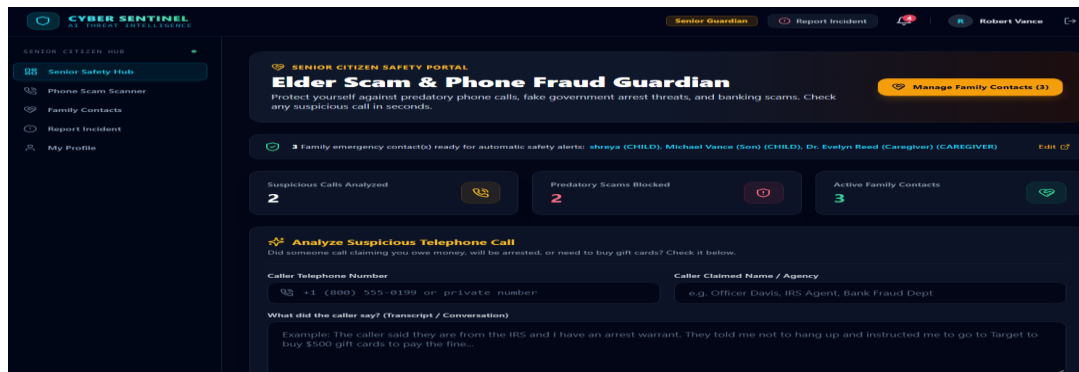
## Registered User – URL Threat Analysis Dashboard



## Cybersecurity Analyst – Incident & Case Management Dashboard



## Senior Citizen User – Phone Call Scam Detection



E-resource :

<https://github.com/shreya080527/Cyber-sentinel-ai-system>